Note

Lab05 Exercise

Advisor：Prof. Chen-Yi Lee

TA：Bang-Yuan Xiao

Date：2025/10/08

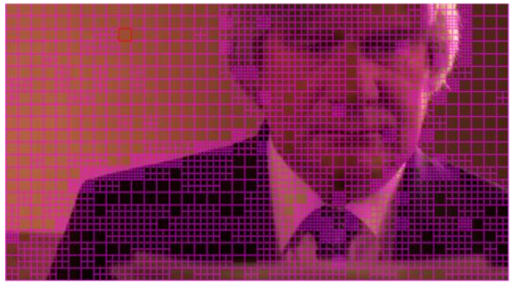
Chapter

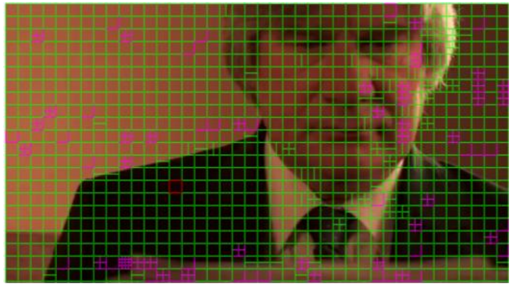
- Overview
- Prediction
- Transform
- Example

Overview

H.264 Lite Prediction and Transform Engine

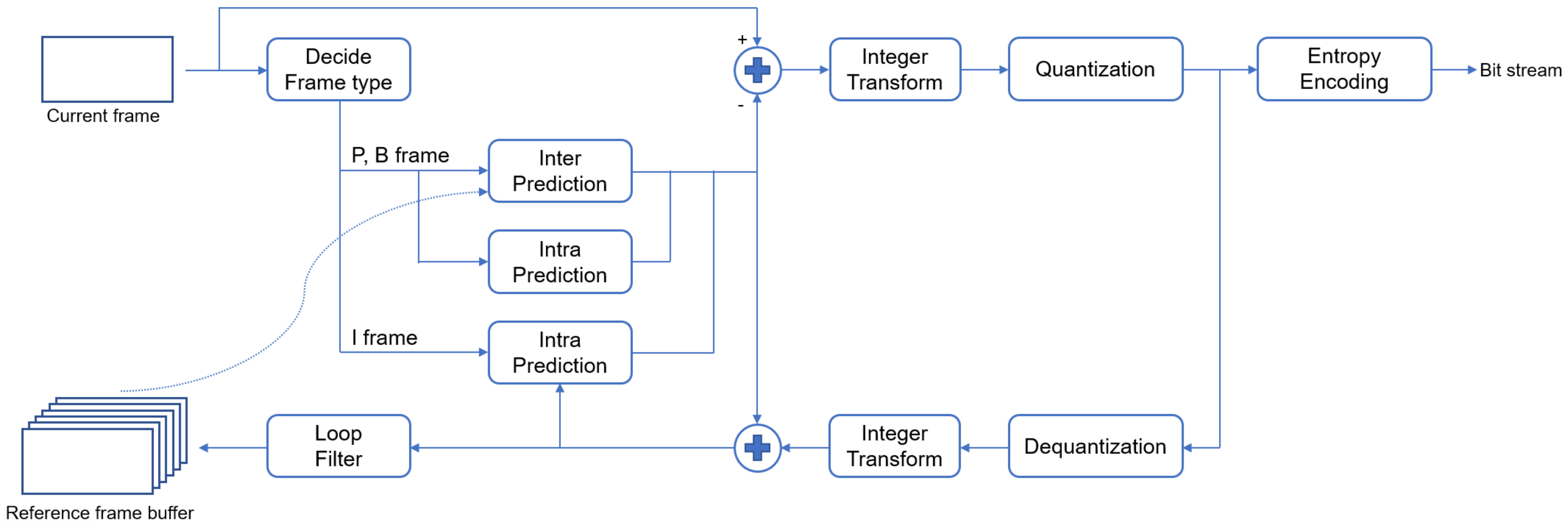
Introduction to H.264


Intra Prediction


Inter Prediction

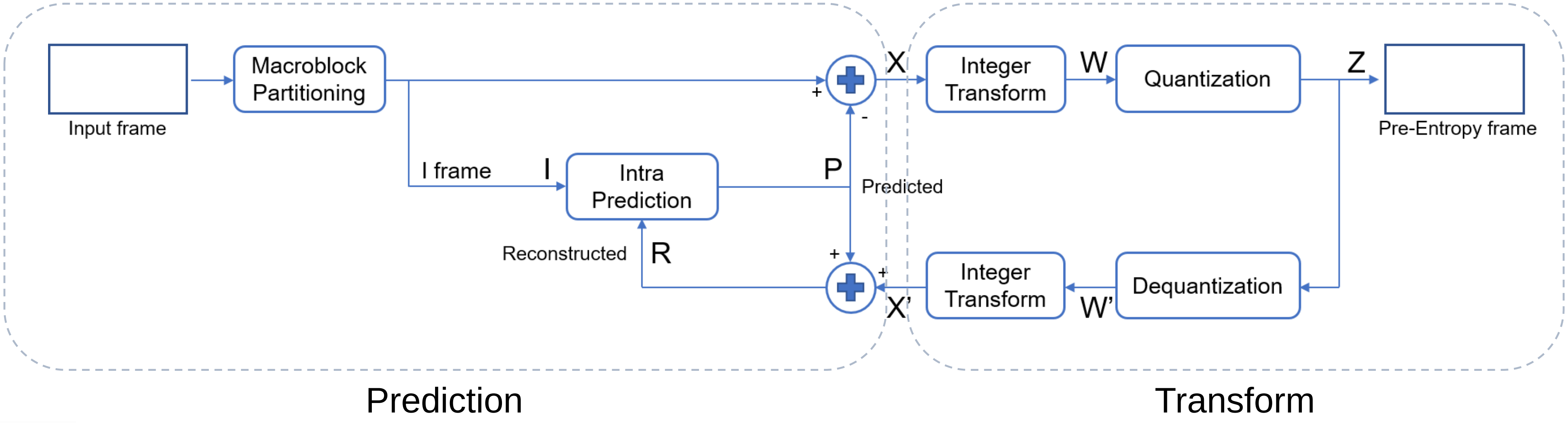
- Prediction
  - Intra prediction (Spatial redundancy)
  - Inter prediction (Temporal redundancy)
  - Obtain residual data
- Transform
  - Concentrate most of the residual energy into low-frequency coefficients
  - Quantization for removing insignificant high-frequency coefficients

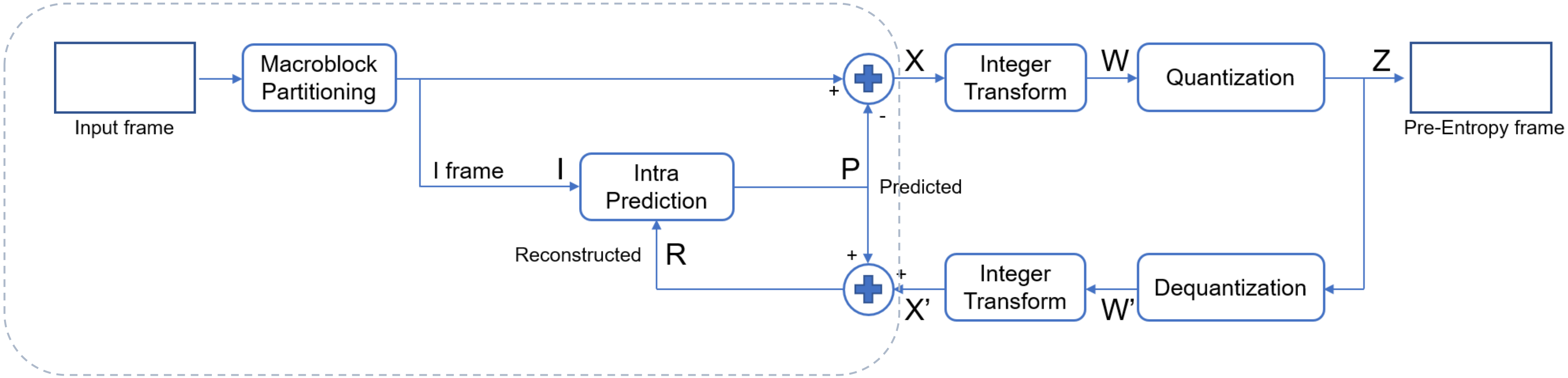
Algorithm for H.264



Algorithm for H.264 Lite

- This lab: H.264 Lite Prediction and Transform Engine (HLPTE)




Input frame
Macroblock Partitioning
I frame
I
Intra Prediction
P
Predicted
Reconstructed
R
+
-
+
+
X
Integer Transform
W
Quantization
Z
Pre-Entropy frame
X'
Integer Transform
W'
Dequantization

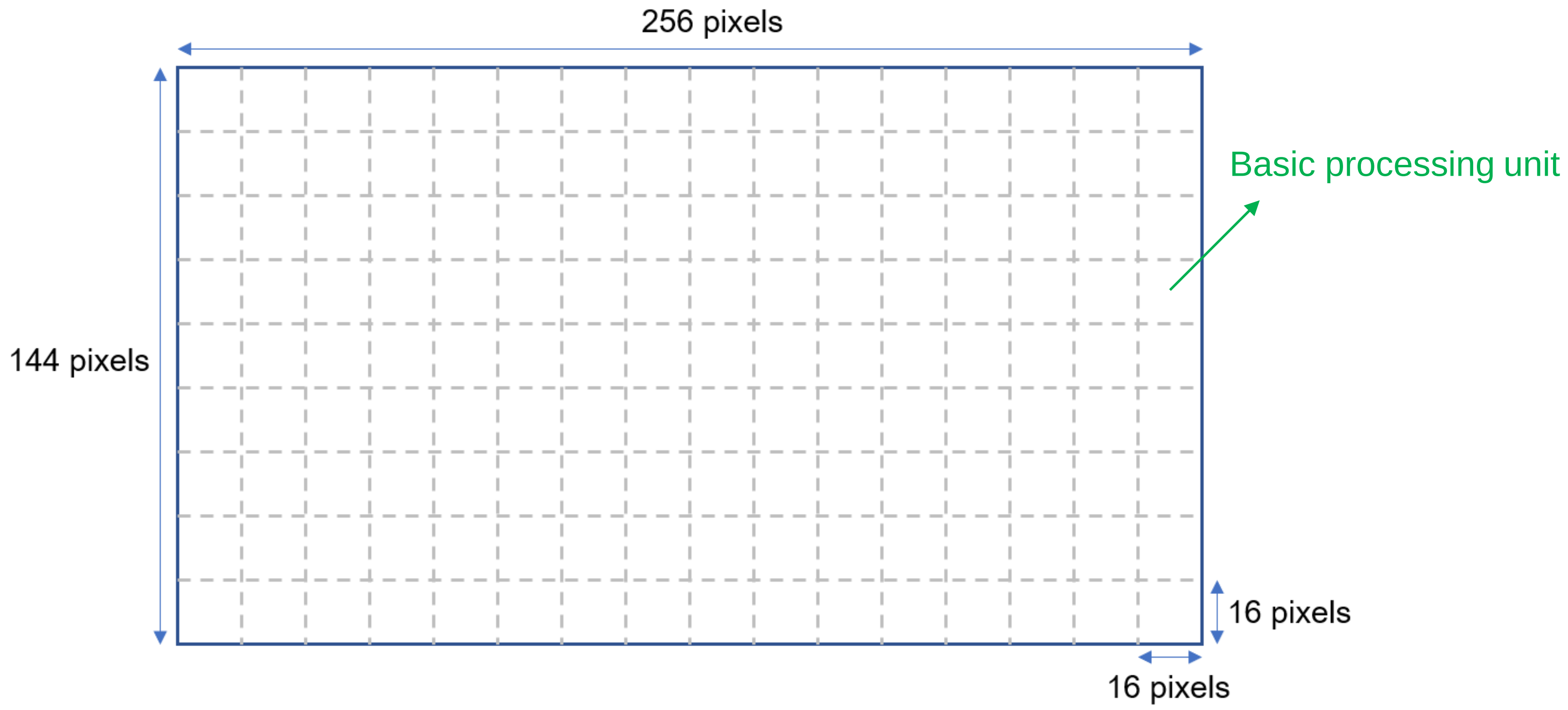
Prediction

Outline

- Macroblock partitioning
- Intra Prediction
- Residual Computation

Macroblock partitioning

- Input frame will be partitioned to several 16x16 macroblocks

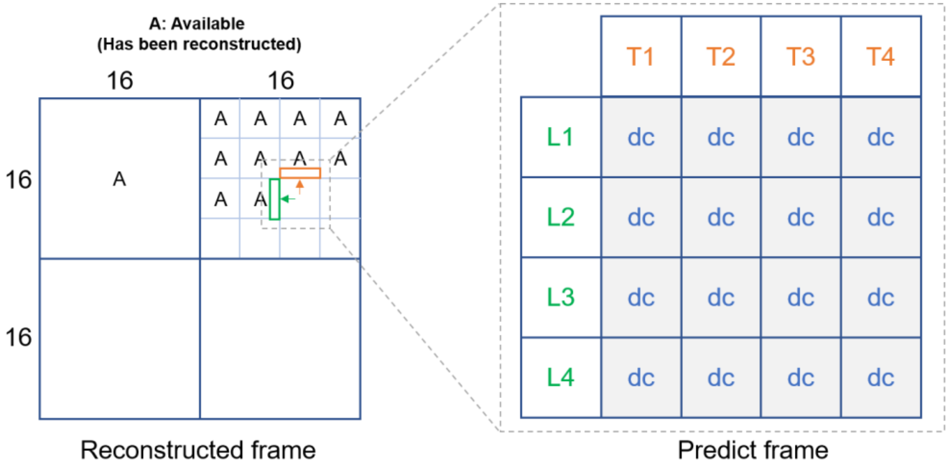


Intra Prediction (1/5)

- DC mode

Reference top pixels: T1, T2, T3, T4

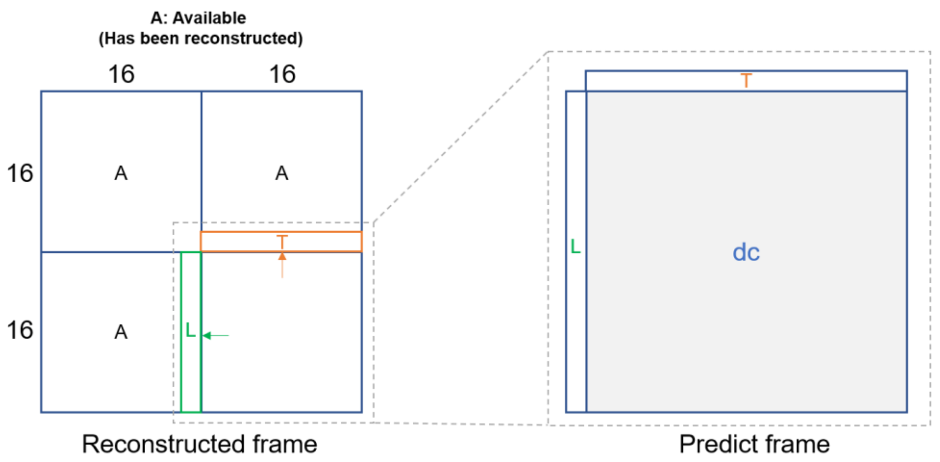
Reference left pixels: L1, L2, L3, L4



$$Predict_{ij} = dc$$

$$dc = \begin{cases} sum(T+L) \gg 3 & if\ both\ T\ and\ L\ are\ available \\ sum(T) \gg 2 & if\ only\ T\ are\ available \\ sum(L) \gg 2 & if\ only\ L\ are\ available \\ 128 & if\ neither\ T\ nor\ L\ are\ available \end{cases}$$

Intra 4x4 Prediction



$$Predict_{ij} = dc$$

$$dc = \begin{cases} sum(T+L) \gg 5 & if\ both\ T\ and\ L\ are\ available \\ sum(T) \gg 4 & if\ only\ T\ are\ available \\ sum(L) \gg 4 & if\ only\ L\ are\ available \\ 128 & if\ neither\ T\ nor\ L\ are\ available \end{cases}$$

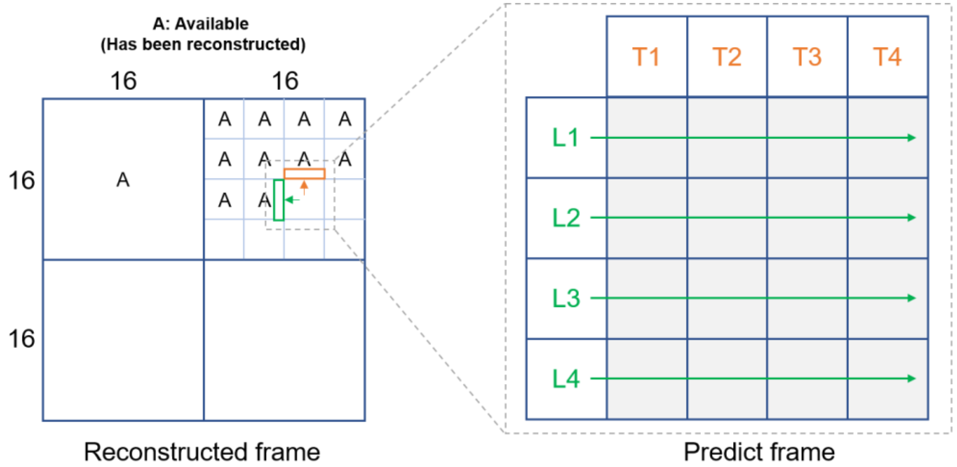
Intra 16x16 Prediction

Intra Prediction (2/5)

- Horizontal mode

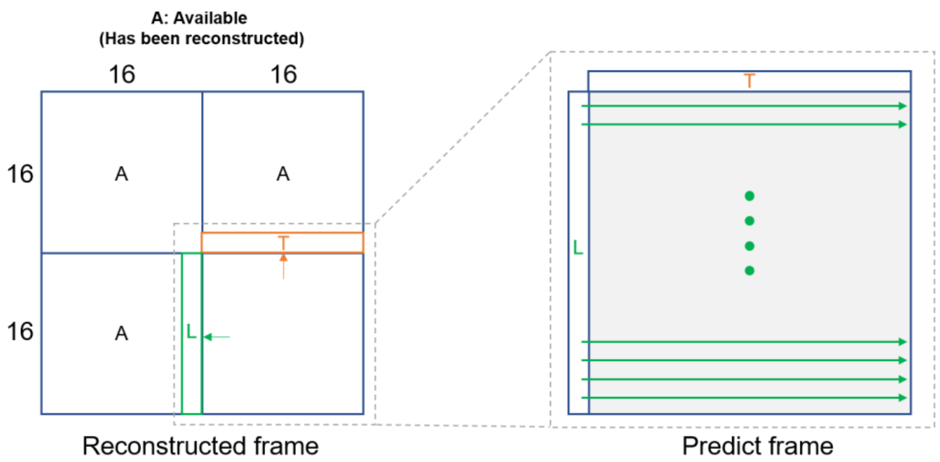
Reference top pixels: T1, T2, T3, T4

Reference left pixels: L1, L2, L3, L4



$Predict_{ij} = L_i$

Intra 4x4 Prediction



$Predict_{ij} = L_i$

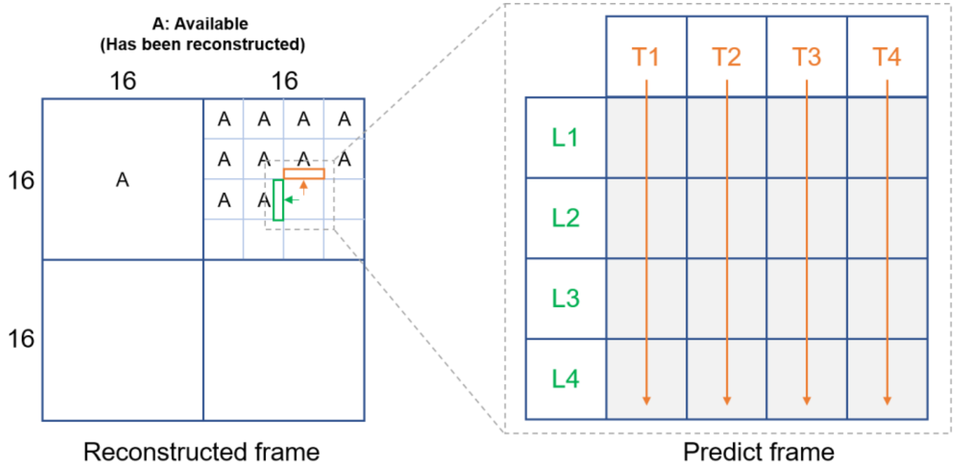
Intra 16x16 Prediction

Intra Prediction (3/5)

- Vertical mode

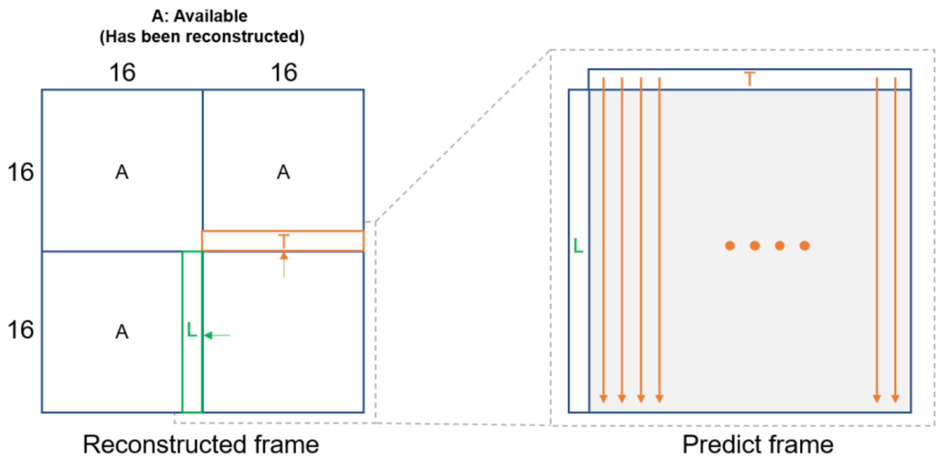
Reference top pixels: T1, T2, T3, T4

Reference left pixels: L1, L2, L3, L4



$Predict_{ij} = T_j$

Intra 4x4 Prediction



$Predict_{ij} = T_j$

Intra 16x16 Prediction

Intra Prediction (4/5)

- Compute the sum of absolute difference (SAD)

$$\mathrm{SAD} = \sum_{i=0}^{3}\sum_{i=0}^{3}|input(i,j) - predict(i,j)|$$

Intra 4x4 Prediction

$$\mathrm{SAD} = \sum_{i=0}^{15}\sum_{i=0}^{15}|input(i,j) - predict(i,j)|$$

Intra 16x16 Prediction

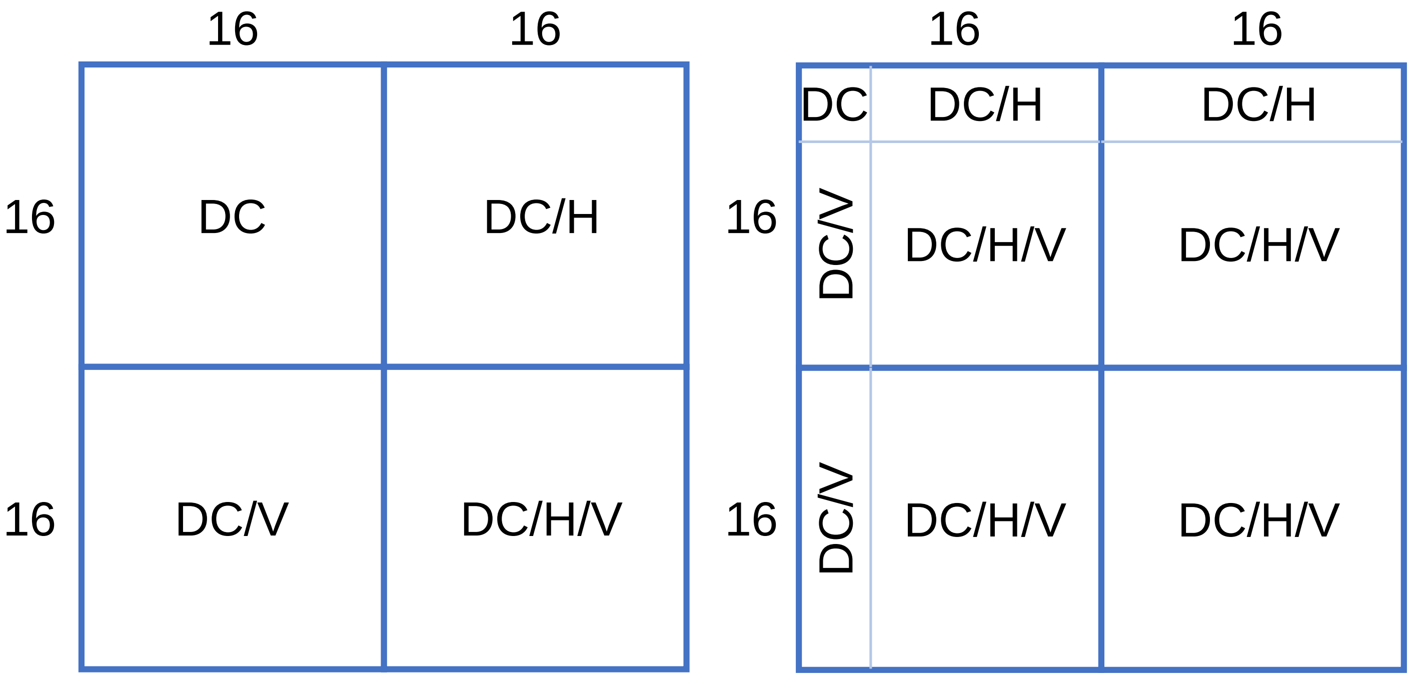
Intra Prediction (5/5)

If two or more modes yield the same SAD,
select the mode according to the priority: DC > Horizontal > Vertical

- How to determine predicted block
  - Select the mode with minimum SAD to predict the frame

1. If **both left and top pixels are available**, evaluate **DC**, **Horizontal**, and **Vertical** modes.
2. If **only left pixels are available**, evaluate **DC** and **Horizontal** modes.
3. If **only top pixels are available**, evaluate **DC** and **Vertical** modes.
4. If **neither left nor top pixels are available**, evaluate **only DC** mode.

summary



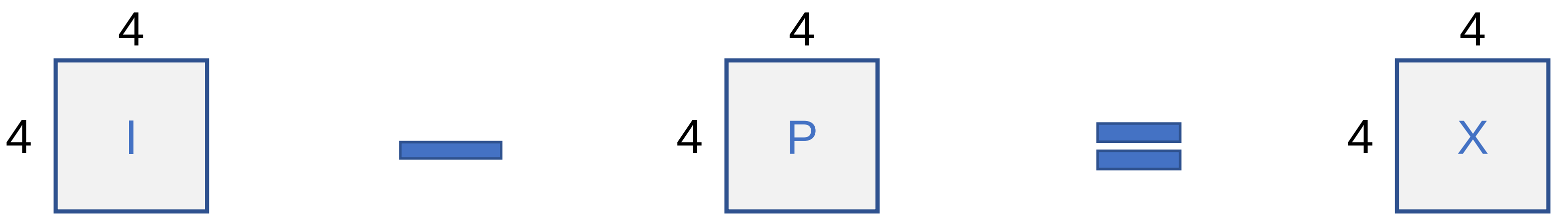
Intra 16x16 prediction

Intra 4x4 prediction

Residual Computation (1/2)

- Case 1: Intra 4x4

$$Residual(i,j) = Input(i,j) - Predicted(i,j)\ for\ each\ pixel$$



4x4 Input block (I) 4x4 Predicted block (P) 4x4 Residual block (X)

Residual Computation (2/2)

- Case 2: Intra 16x16

$$Residual(i,j) = Input(i,j) - Predicted(i,j) \; for \; each \; pixel$$

16 × 16

I_{00}	I_{01}	I_{02}	I_{03}
I_{10}	I_{11}	I_{12}	I_{13}
I_{20}	I_{21}	I_{22}	I_{23}
I_{30}	I_{31}	I_{32}	I_{33}

16x16 Input block (I)

$-$

16 × 16

P_{00}	P_{01}	P_{02}	P_{03}
P_{10}	P_{11}	P_{12}	P_{13}
P_{20}	P_{21}	P_{22}	P_{23}
P_{30}	P_{31}	P_{32}	P_{33}

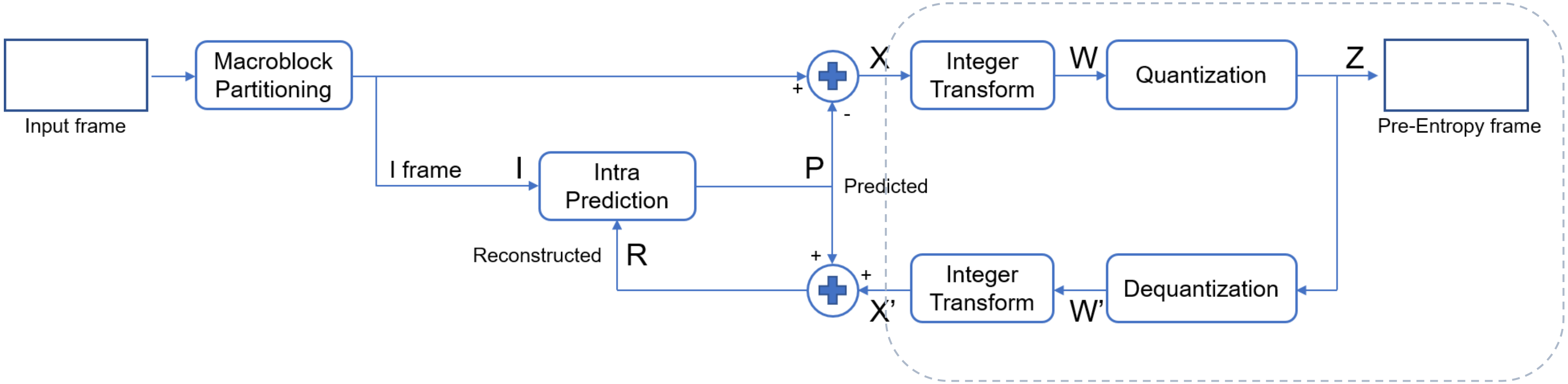
16x16 Predicted block (P)

$=$

16 × 16

X_{00}	X_{01}	X_{02}	X_{03}
X_{10}	X_{11}	X_{12}	X_{13}
X_{20}	X_{21}	X_{22}	X_{23}
X_{30}	X_{31}	X_{32}	X_{33}

16x16 Residual block (X)


Input frame
Macroblock Partitioning
I frame
I
Intra Prediction
P
Predicted
Reconstructed
R
+
-
X
Integer Transform
W
Quantization
Z
Pre-Entropy frame
+
+
X'
Integer Transform
W'
Dequantization

Transform

Outline

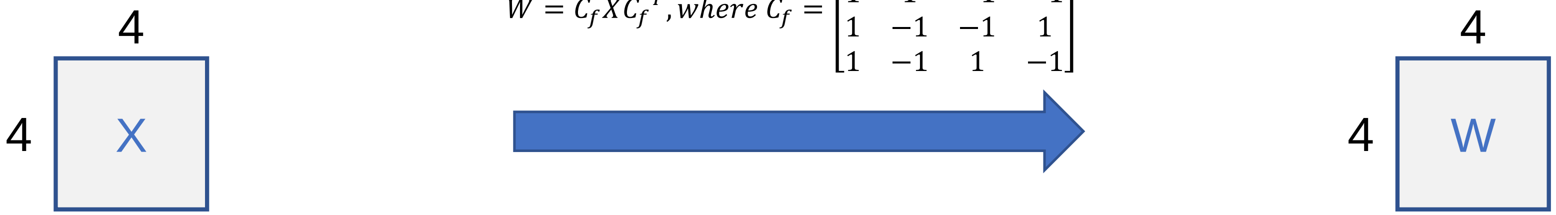
- Integer Transform
- Quantization
- Dequantization
- Inverse Integer Transform
- Reconstruction

Integer Transform (1/2)

- Case 1: Intra 4x4

X is represented as 4x4 residual block

$$W = C_f X C_f^{\,T}, where\ C_f = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$



4x4 Residual block (X)

4x4 Transformed block (W)

Integer Transform (2/2)

- Case 2: Intra 16x16

16x16 Residual block (X) — 16 × 16

X_{00}	X_{01}	X_{02}	X_{03}
X_{10}	X_{11}	X_{12}	X_{13}
X_{20}	X_{21}	X_{22}	X_{23}
X_{30}	X_{31}	X_{32}	X_{33}

X is represented as 4x4 residual block

$$W = C_f X C_f^{\,T}, where\ C_f = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

16x16 Transformed block (W) — 16 × 16

W_{00}	W_{01}	W_{02}	W_{03}
W_{10}	W_{11}	W_{12}	W_{13}
W_{20}	W_{21}	W_{22}	W_{23}
W_{30}	W_{31}	W_{32}	W_{33}

Quantization (1/2)

- Case 1: Intra 4x4

Multiplication factor MF

QP mod 6	a	b	c
0	13107	5243	8066
1	11916	4660	7490
2	10082	4194	6554
3	9362	3647	5825
4	8192	3355	5243
5	7282	2893	4559

Quantization offset

QP	f
0~5	10922
6~11	21845
12~17	43690
18~23	87381
24~29	174762

$$qbits = 15 + floor\left(\frac{QP}{6}\right)$$

The value and sign of Zij are calculated independently

$$|Z_{ij}| = (|W_{ij}| \cdot MF_{ij} + f) \gg qbits$$

$$sign(Z_{ij}) = sign(W_{ij})$$

$$\text{MF} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

4 × 4 W — 4x4 Transformed block (W)

4 × 4 Z — 4x4 Pre-entropy block (Z)

If W_{00} = -200, MF_{00} = 7282, f = 174762, qbits = 19, then
1. The value of Z_{00} = (200*7282+174762) >> 19 = 3
2. The sign of Z_{00} = - since W_{00} < 0
Therefore, Z_{00} = -3

Quantization (2/2)

- Case 2: Intra 16x16

Multiplication factor MF

QP mod 6	a	b	c
0	13107	5243	8066
1	11916	4660	7490
2	10082	4194	6554
3	9362	3647	5825
4	8192	3355	5243
5	7282	2893	4559

Quantization offset

QP	f
0~5	10922
6~11	21845
12~17	43690
18~23	87381
24~29	174762

$$qbits = 15 + floor\left(\frac{QP}{6}\right)$$

16 × 16

W_{00}	W_{01}	W_{02}	W_{03}
W_{10}	W_{11}	W_{12}	W_{13}
W_{20}	W_{21}	W_{22}	W_{23}
W_{30}	W_{31}	W_{32}	W_{33}

16x16 Transformed block (W)

$$|Z_{ij}| = \left(|W_{ij}| \cdot MF_{ij} + f\right) \gg qbits$$

$$sign(Z_{ij}) = sign(W_{ij})$$

$$\mathrm{MF} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

16 × 16

Z_{00}	Z_{01}	Z_{02}	Z_{03}
Z_{10}	Z_{11}	Z_{12}	Z_{13}
Z_{20}	Z_{21}	Z_{22}	Z_{23}
Z_{30}	Z_{31}	Z_{32}	Z_{33}

16x16 Pre-entropy block (Z)

Dequantization (1/2)

- Case 1: Intra 4x4

Scaling factor V

QP mod 6	a	b	c
0	10	16	13
1	11	18	14
2	13	20	16
3	14	23	18
4	16	25	20
5	18	29	23

4

4 Z

4x4 Pre-entropy block (Z)

$$W'_{ij} = Z_{ij} \cdot V_{ij} \cdot 2^{floor(QP/6)}$$

$$\mathrm{V} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

4

4 W’

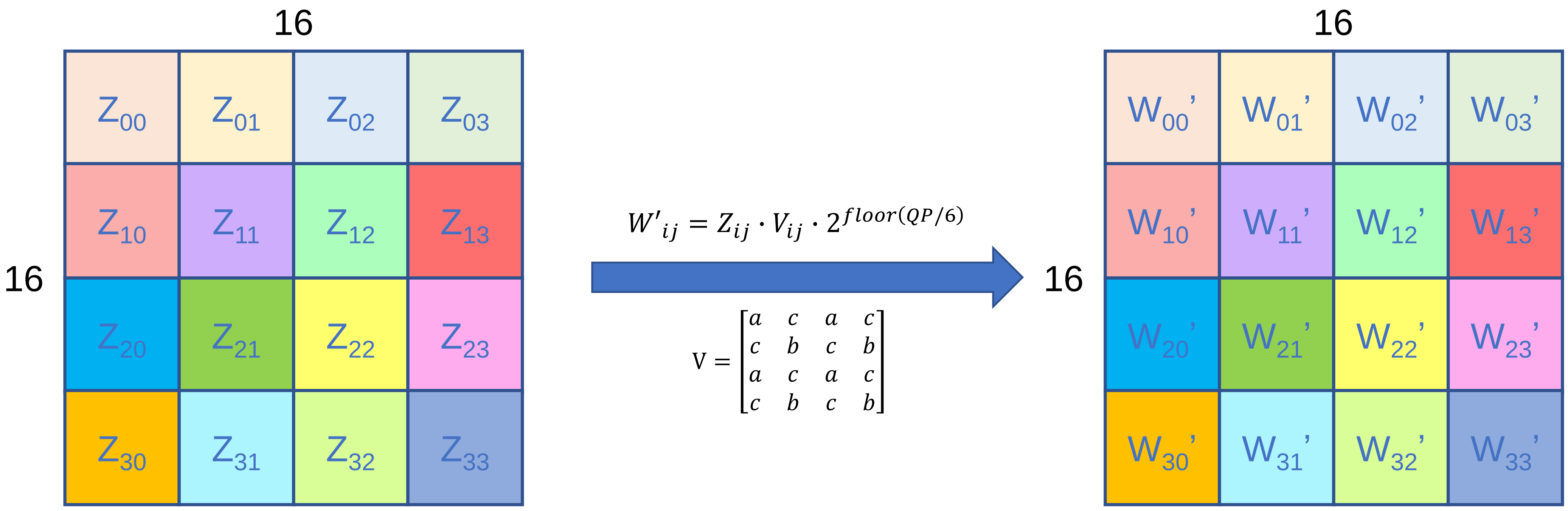
4x4 Dequantized block (W’)

Dequantization (2/2)

- Case 2: Intra 16x16

Scaling factor V

QP mod 6	a	b	c
0	10	16	13
1	11	18	14
2	13	20	16
3	14	23	18
4	16	25	20
5	18	29	23



16x16 Pre-entropy block (Z)

16x16 Dequantized block (W')

Inverse Integer Transform (1/2)

- Case 1: Intra 4x4

$$Y = C_i^{\,T} W' C_i, where\ C_i = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$X' = Y >>> 6$$

Original formula: $X' = Y/64$

4

4 W'

4x4 Dequantized block (W')

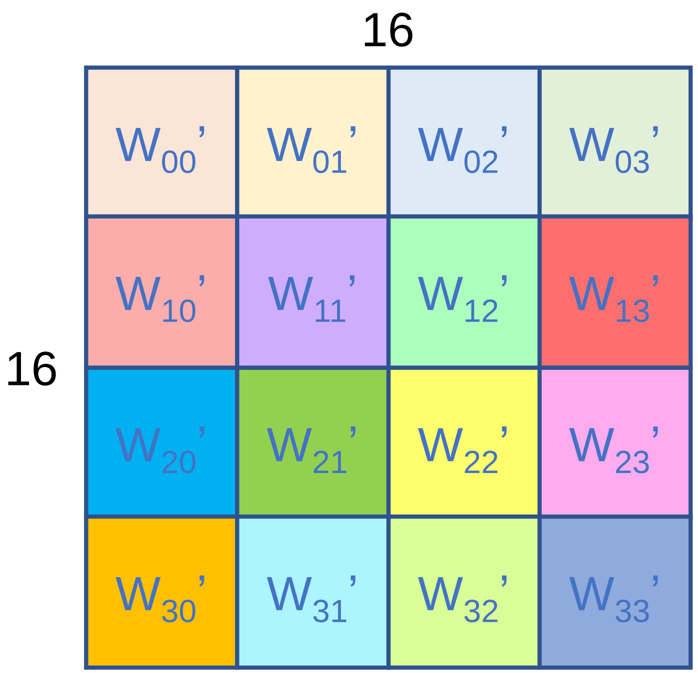
4

4 X'

4x4 inverse-transformed block (X')

Inverse Integer Transform (2/2)

- Case 2: Intra 16x16

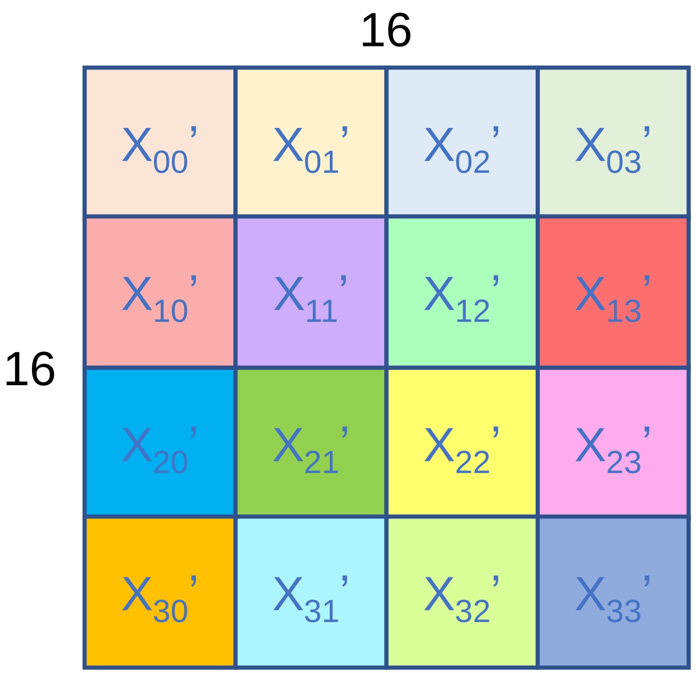


16x16 Dequantized block (W')

$$Y = C_i^T W' C_i, where\ C_i = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

$$X' = Y >>> 6$$

Original formula: $X' = Y/64$



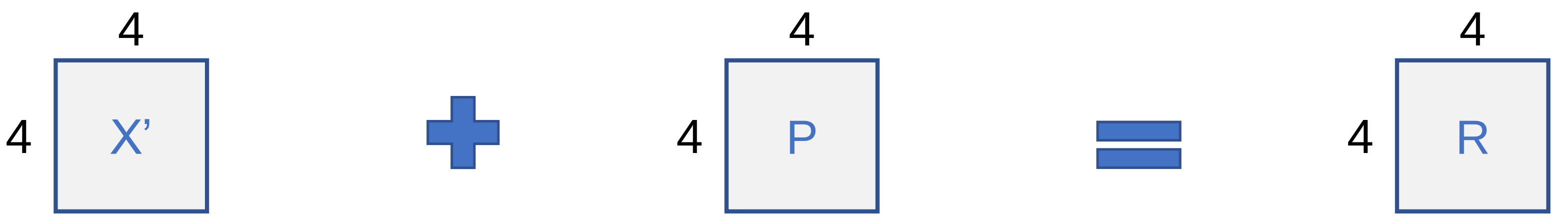
16x16 inverse-transformed block (X')

Reconstruction (1/2)

- Case 1: Intra 4x4

Note: If Reconstructed (i, j) exceeds 255, set it to 255. If Reconstructed (i, j) is less than 0, set it to 0.

$$Reconstructed(i,j) = X'(i,j) + Predicted(i,j)\ for\ each\ pixel$$



4x4 Inverse-transformed block (X’) 4x4 Predicted block (P) 4x4 Reconstructed block (R)

Reconstruction (2/2)

- Case 2: Intra 16x16

Note: If Reconstructed (i, j) exceeds 255, set it to 255. If Reconstructed (i, j) is less than 0, set it to 0.

$$Reconstructed(i,j) = X'(i,j) + Predicted(i,j)\ for\ each\ pixel$$

16 × 16

X_{00}'	X_{01}'	X_{02}'	X_{03}'
X_{10}'	X_{11}'	X_{12}'	X_{13}'
X_{20}'	X_{21}'	X_{22}'	X_{23}'
X_{30}'	X_{31}'	X_{32}'	X_{33}'

16x16 inverse-transformed block

+

16 × 16

P_{00}	P_{01}	P_{02}	P_{03}
P_{10}	P_{11}	P_{12}	P_{13}
P_{20}	P_{21}	P_{22}	P_{23}
P_{30}	P_{31}	P_{32}	P_{33}

16x16 Prediction block

=

16 × 16

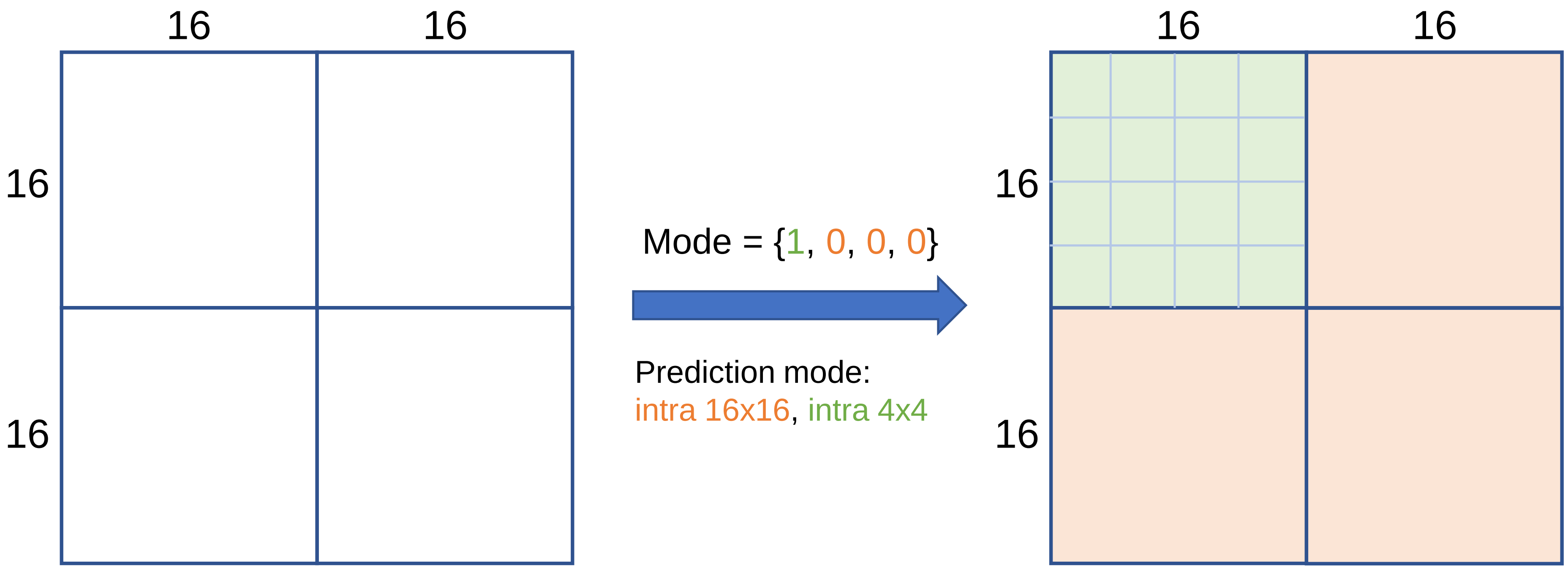
R_{00}	R_{01}	R_{02}	R_{03}
R_{10}	R_{11}	R_{12}	R_{13}
R_{20}	R_{21}	R_{22}	R_{23}
R_{30}	R_{31}	R_{32}	R_{33}

16x16 Reconstructed block

Example

Case 1 – Intra 4x4 (1/15)

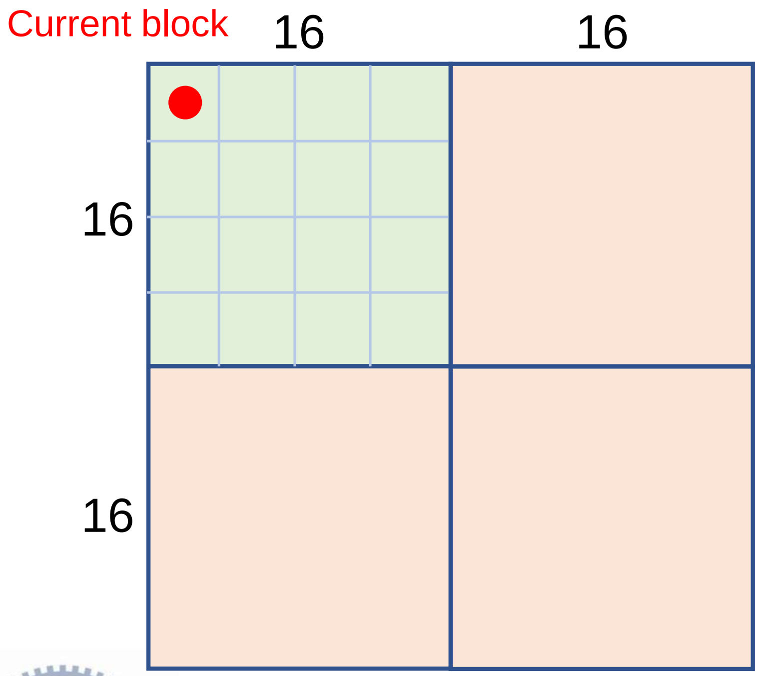
- Fetch the 32x32 frame based on index
- Mode: {1, 0, 0, 0}



Case 1 – Intra 4x4 (2/15)

- Round 1: Intra prediction

Select the mode with the minimum SAD!



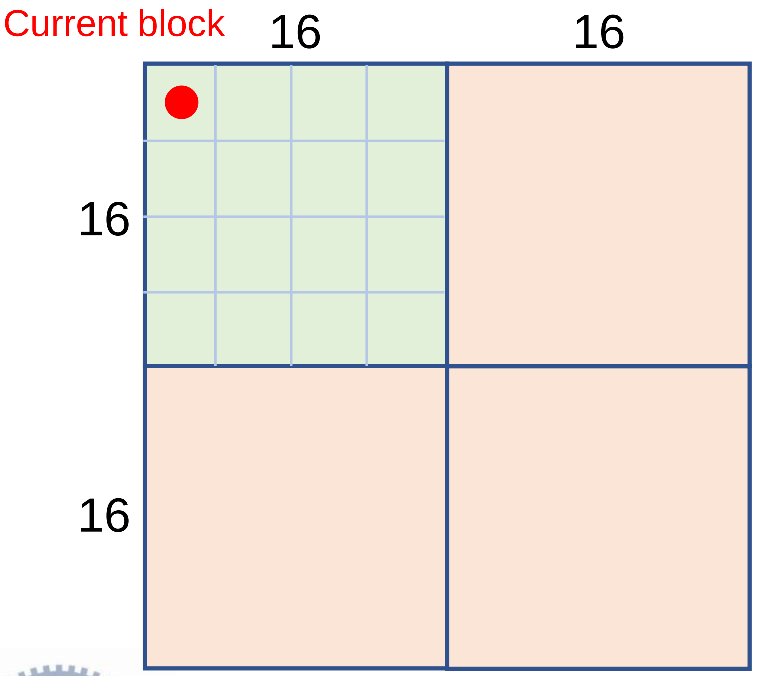
Input frame

	N/A	N/A	N/A	N/A
N/A	128	128	128	128
N/A	128	128	128	128
N/A	128	128	128	128
N/A	128	128	128	128

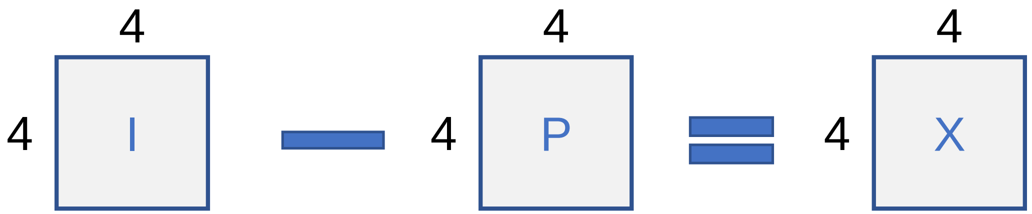
4x4 Predicted block (DC)

Case 1 – Intra 4x4 (3/15)

- Round 1: Residual computation

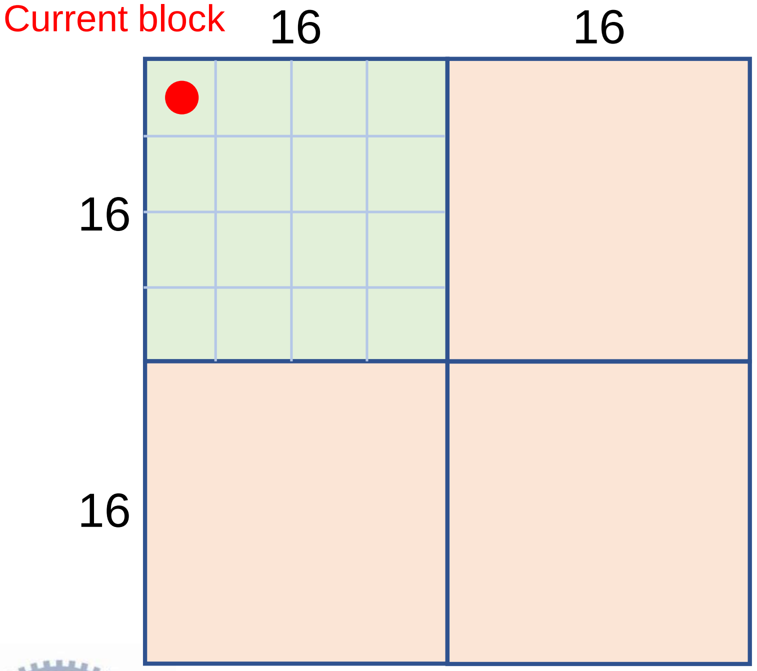


Input frame



Case 1 – Intra 4x4 (4/15)

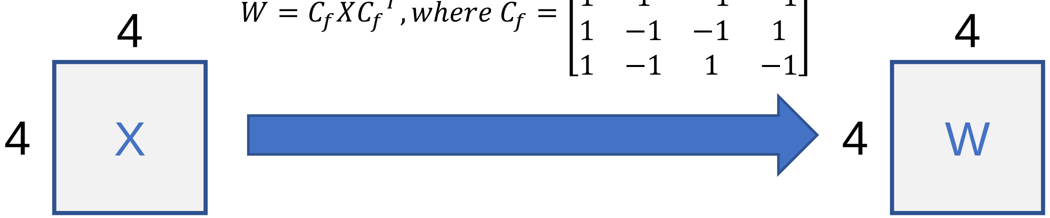
- Round 1: Integer Transform



Input frame

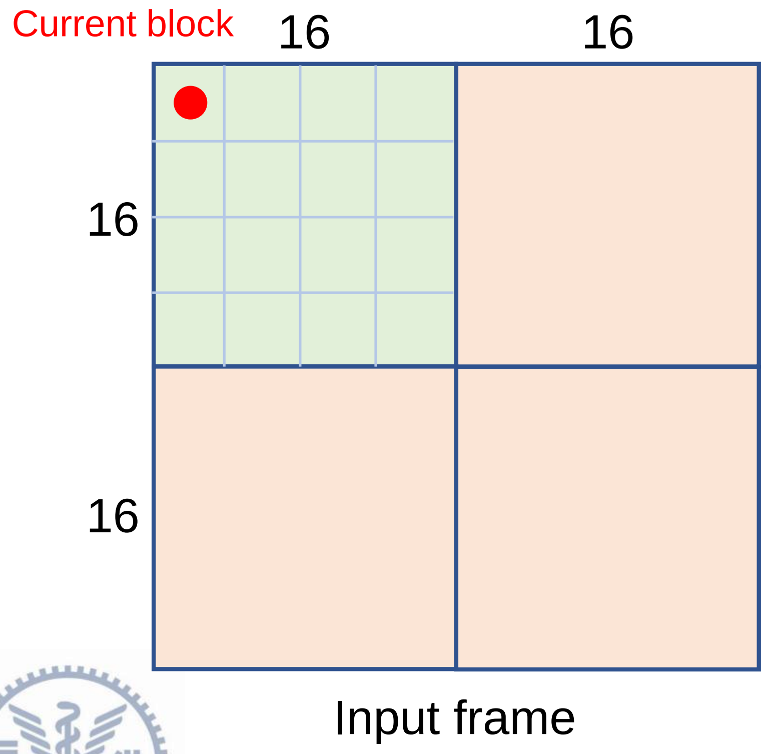
X is represented as 4x4 residual block

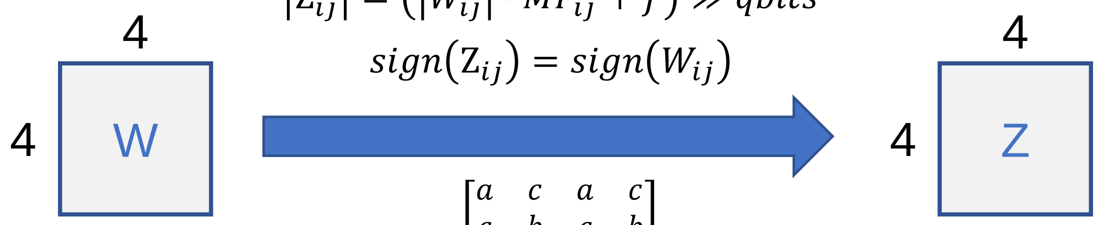
$$W = C_f X C_f^{\,T}, where\ C_f = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$



Case 1 – Intra 4x4 (5/15)

- Round 1: Quantization





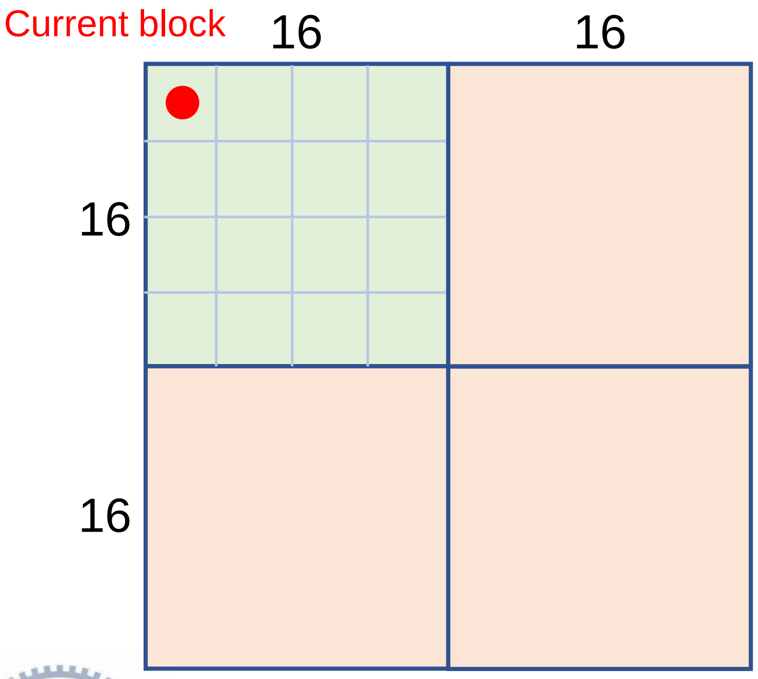
$$|Z_{ij}| = (|W_{ij}| \cdot MF_{ij} + f) \gg qbits$$

$$sign(Z_{ij}) = sign(W_{ij})$$

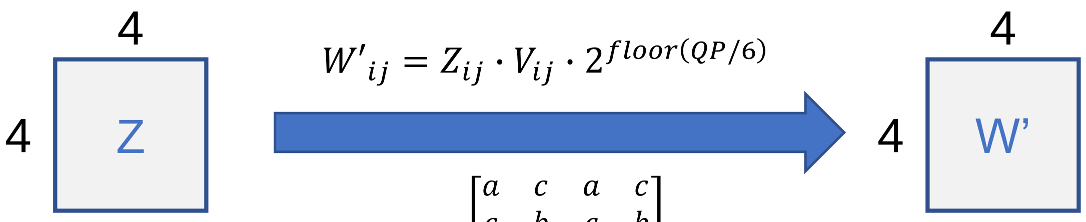
$$\mathrm{MF} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

Case 1 – Intra 4x4 (6/15)

- Round 1: Dequantization



Input frame

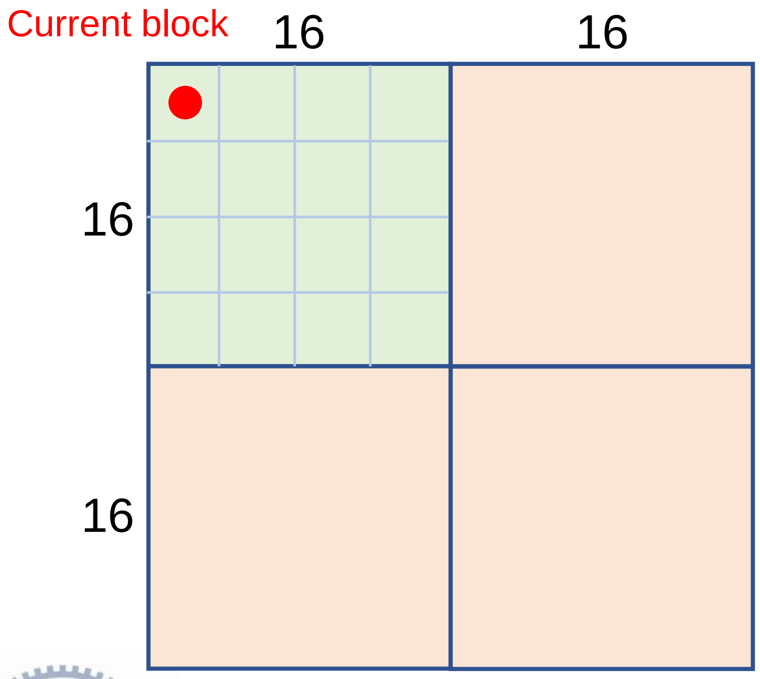


$$W'_{ij} = Z_{ij} \cdot V_{ij} \cdot 2^{floor(QP/6)}$$

$$\mathrm{V} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

Case 1 – Intra 4x4 (7/15)

- Round 1: Inverse Integer Transform

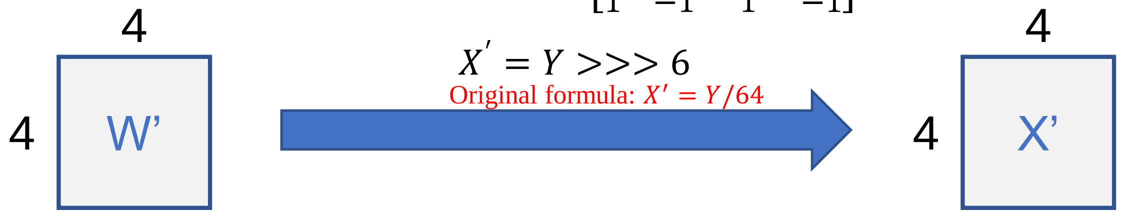


Input frame

$$Y = C_i^{\,T} W' C_i, where\ C_i = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

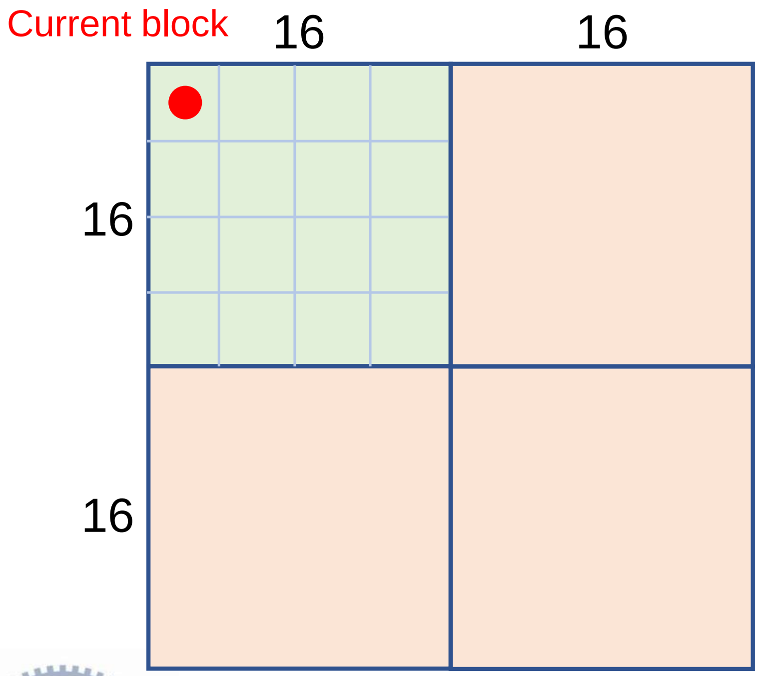
$$X' = Y >>> 6$$

Original formula: $X' = Y/64$

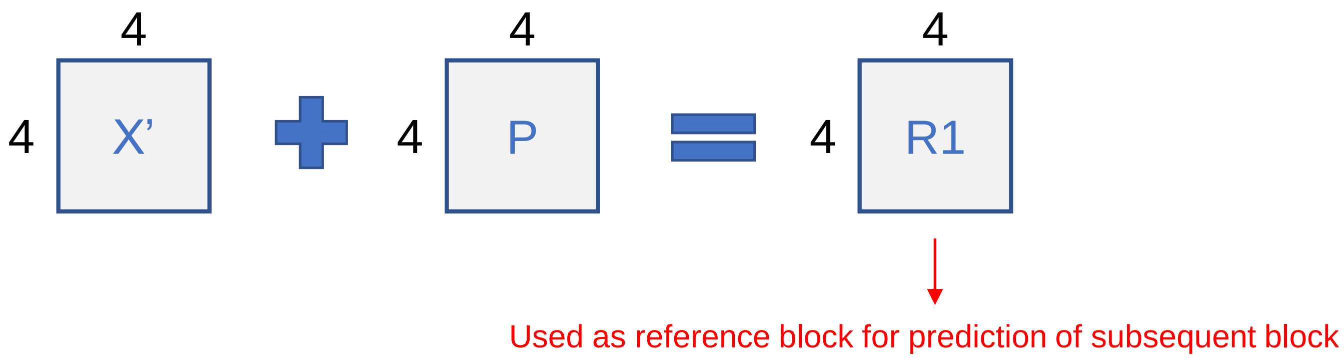


Case 1 – Intra 4x4 (8/15)

- Round 1: Reconstruction



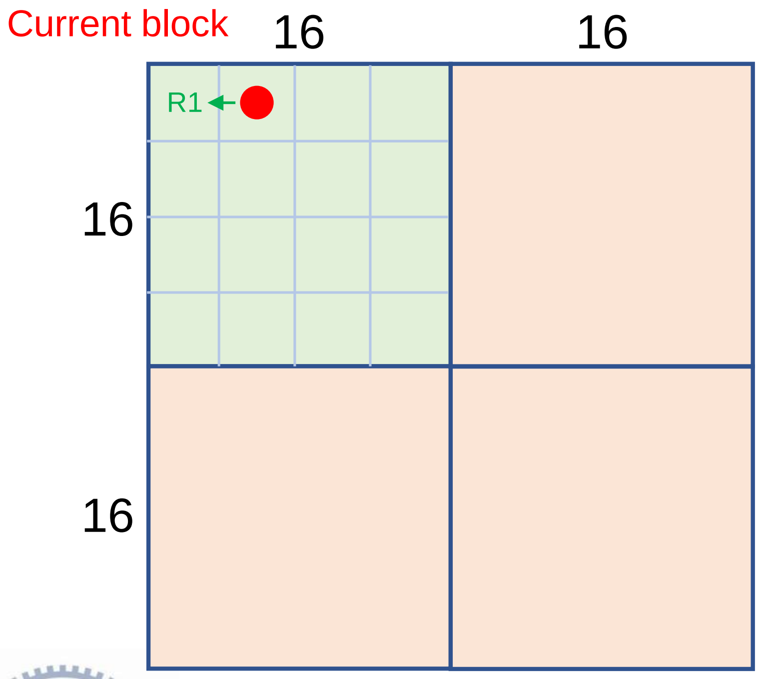
Input frame



Case 1 – Intra 4x4 (9/15)

- Round 2: Intra Prediction

Select the mode with the minimum SAD!



Input frame

dc = ($R1_{(0,3)}$+$R1_{(1,3)}$+$R1_{(2,3)}$+$R1_{(3,3)}$) >> 2

	N/A	N/A	N/A	N/A
$R1_{(0,3)}$	dc	dc	dc	dc
$R1_{(1,3)}$	dc	dc	dc	dc
$R1_{(2,3)}$	dc	dc	dc	dc
$R1_{(3,3)}$	dc	dc	dc	dc

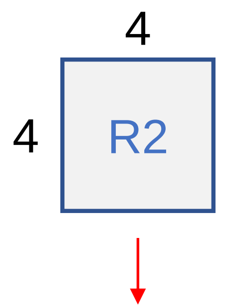
4x4 Predicted block (DC)

	N/A	N/A	N/A	N/A
$R1_{(0,3)}$	$R1_{(0,3)}$	$R1_{(0,3)}$	$R1_{(0,3)}$	$R1_{(0,3)}$
$R1_{(1,3)}$	$R1_{(1,3)}$	$R1_{(1,3)}$	$R1_{(1,3)}$	$R1_{(1,3)}$
$R1_{(2,3)}$	$R1_{(2,3)}$	$R1_{(2,3)}$	$R1_{(2,3)}$	$R1_{(2,3)}$
$R1_{(3,3)}$	$R1_{(3,3)}$	$R1_{(3,3)}$	$R1_{(3,3)}$	$R1_{(3,3)}$

4x4 Predicted block (Horizontal)

Case 1 – Intra 4x4 (10/15)

- Round 2
  - Residual computation
  - Integer Transform
  - Quantization
  - Dequantization
  - Inverse Integer Transform
  - Reconstruction

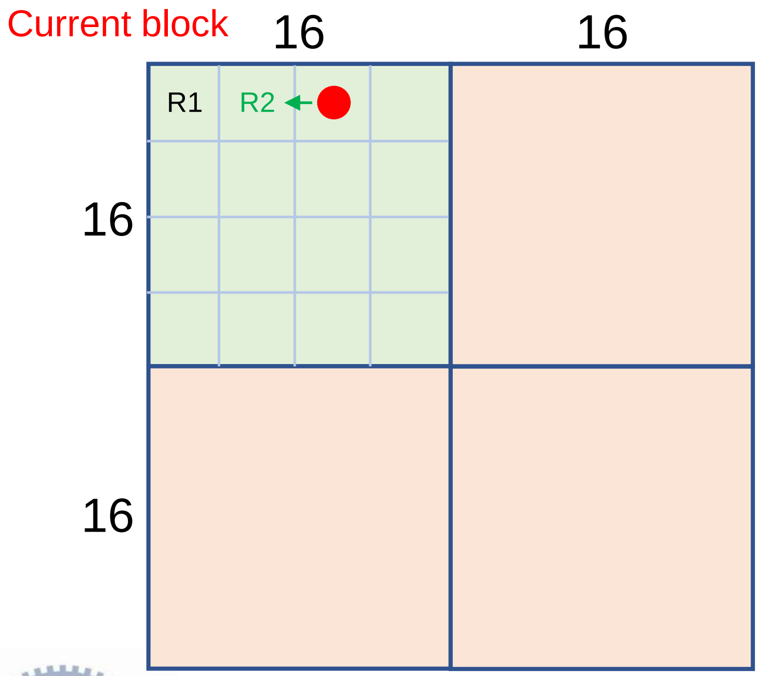


Used as reference block for prediction of subsequent block

Case 1 – Intra 4x4 (11/15)

- Round 3: Intra Prediction

Select the mode with the minimum SAD!



Input frame

$dc = (R2_{(0,3)}+R2_{(1,3)}+R2_{(2,3)}+R2_{(3,3)}) >> 2$

	N/A	N/A	N/A	N/A
R2 (0,3)	dc	dc	dc	dc
R2 (1,3)	dc	dc	dc	dc
R2 (2,3)	dc	dc	dc	dc
R2 (3,3)	dc	dc	dc	dc

4x4 Predicted block (DC)

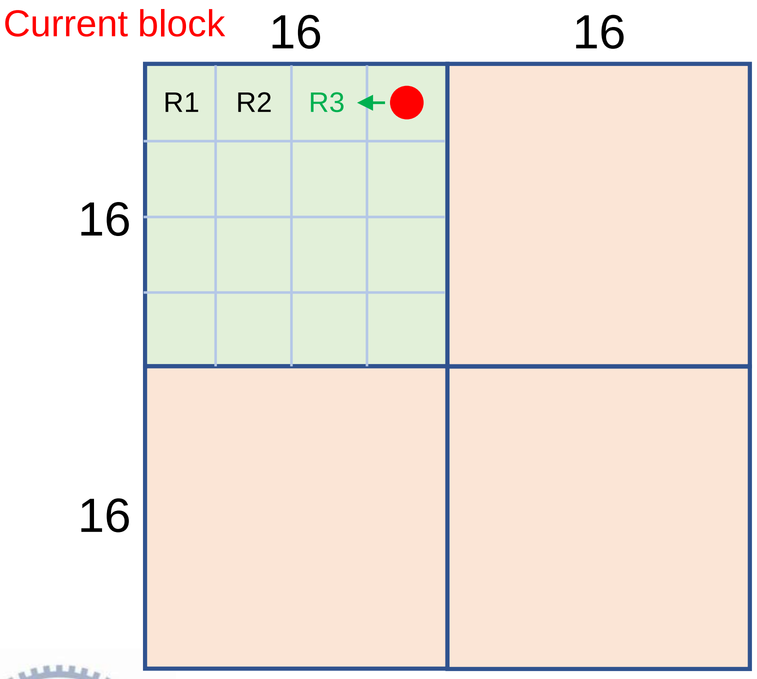
	N/A	N/A	N/A	N/A
R2 (0,3)	R2 (0,3)	R2 (0,3)	R2 (0,3)	R2 (0,3)
R2 (1,3)	R2 (1,3)	R2 (1,3)	R2 (1,3)	R2 (1,3)
R2 (2,3)	R2 (2,3)	R2 (2,3)	R2 (2,3)	R2 (2,3)
R2 (3,3)	R2 (3,3)	R2 (3,3)	R2 (3,3)	R2 (3,3)

4x4 Predicted block (Horizontal)

Case 1 – Intra 4x4 (12/15)

- Round 4: Intra Prediction

Select the mode with the minimum SAD!



Input frame

$dc = (R3_{(0,3)}+R3_{(1,3)}+R3_{(2,3)}+R3_{(3,3)}) >> 2$

	N/A	N/A	N/A	N/A
R3 (0,3)	dc	dc	dc	dc
R3 (1,3)	dc	dc	dc	dc
R3 (2,3)	dc	dc	dc	dc
R3 (3,3)	dc	dc	dc	dc

4x4 Predicted block (DC)

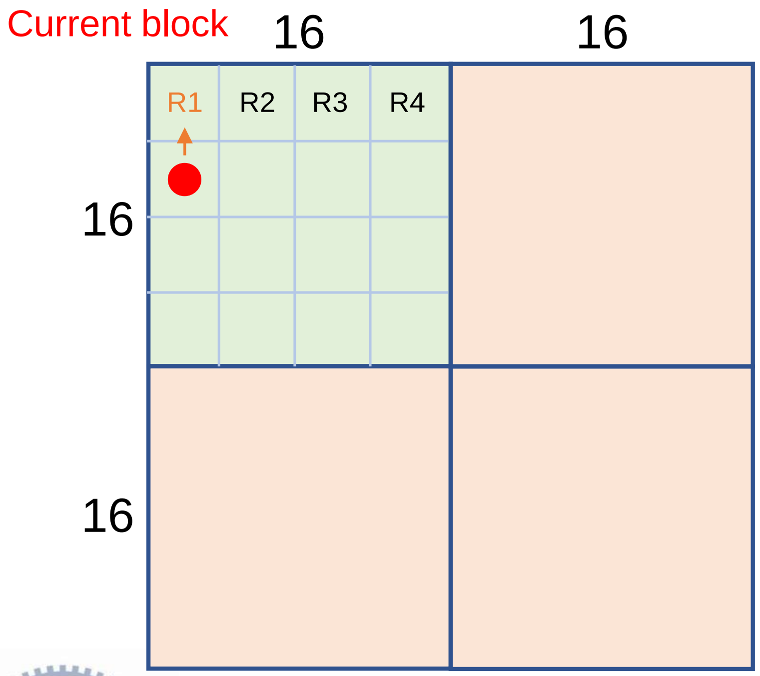
	N/A	N/A	N/A	N/A
R3 (0,3)	R3 (0,3)	R3 (0,3)	R3 (0,3)	R3 (0,3)
R3 (1,3)	R3 (1,3)	R3 (1,3)	R3 (1,3)	R3 (1,3)
R3 (2,3)	R3 (2,3)	R3 (2,3)	R3 (2,3)	R3 (2,3)
R3 (3,3)	R3 (3,3)	R3 (3,3)	R3 (3,3)	R3 (3,3)

4x4 Predicted block (Horizontal)

Case 1 – Intra 4x4 (13/15)

- Round 5: Intra Prediction

Select the mode with the minimum SAD!



Input frame

dc = $(R1_{(3,0)}+R1_{(3,1)}+R1_{(3,2)}+R1_{(3,3)}) >> 2$

	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc

4x4 Predicted block (DC)

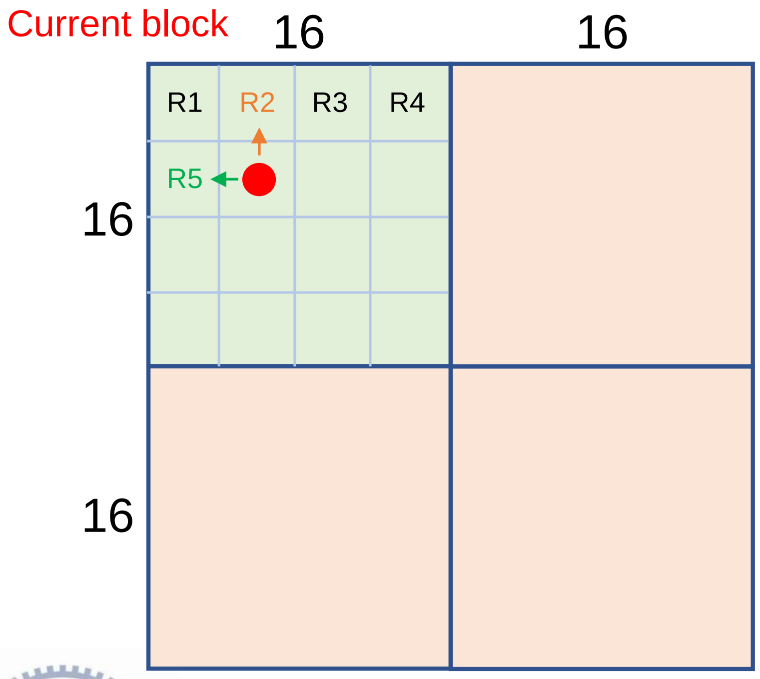
	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)
N/A	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)
N/A	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)
N/A	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)
N/A	R1 (3,0)	R1 (3,1)	R1 (3,2)	R1 (3,3)

4x4 Predicted block (Vertical)

Case 1 – Intra 4x4 (14/15)

- Round 6: Intra Prediction

Select the mode with the minimum SAD!



Input frame

$$dc = (R5_{(0,3)}+R5_{(1,3)}+R5_{(2,3)}+R5_{(3,3)}+R2_{(3,0)}+R2_{(3,1)}+R2_{(3,2)}+R2_{(3,3)}) >> 3$$

	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(0,3)}$	dc	dc	dc	dc
$R5_{(1,3)}$	dc	dc	dc	dc
$R5_{(2,3)}$	dc	dc	dc	dc
$R5_{(3,3)}$	dc	dc	dc	dc

4x4 Predicted block (DC)

	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(0,3)}$	$R5_{(0,3)}$	$R5_{(0,3)}$	$R5_{(0,3)}$	$R5_{(0,3)}$
$R5_{(1,3)}$	$R5_{(1,3)}$	$R5_{(1,3)}$	$R5_{(1,3)}$	$R5_{(1,3)}$
$R5_{(2,3)}$	$R5_{(2,3)}$	$R5_{(2,3)}$	$R5_{(2,3)}$	$R5_{(2,3)}$
$R5_{(3,3)}$	$R5_{(3,3)}$	$R5_{(3,3)}$	$R5_{(3,3)}$	$R5_{(3,3)}$

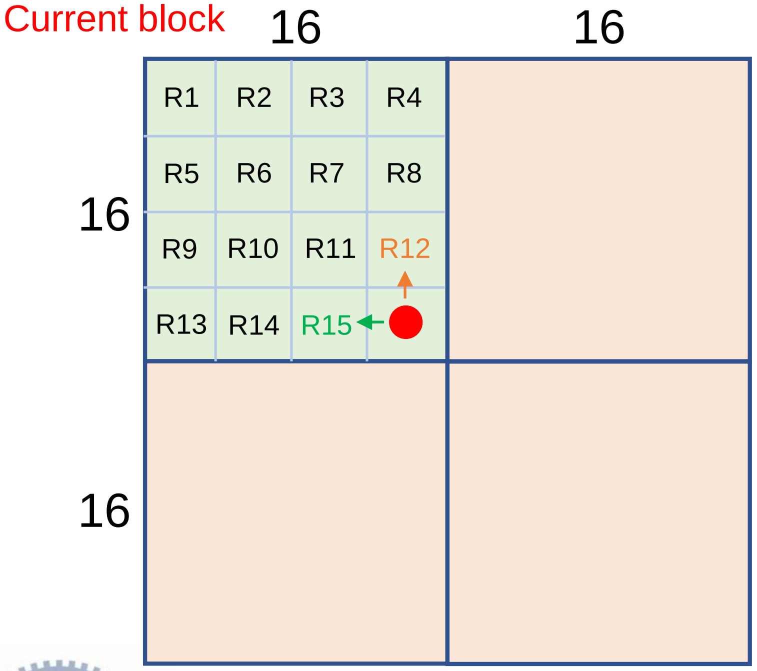
4x4 Predicted block (Horizontal)

	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(0,3)}$	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(1,3)}$	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(2,3)}$	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$
$R5_{(3,3)}$	$R2_{(3,0)}$	$R2_{(3,1)}$	$R2_{(3,2)}$	$R2_{(3,3)}$

4x4 Predicted block (Vertical)

Case 1 – Intra 4x4 (15/15)

- Round 16: Intra Prediction Select the mode with the minimum SAD!



Input frame

dc = ($R15_{(0,3)}$+$R15_{(1,3)}$+$R15_{(2,3)}$+$R15_{(3,3)}$+$R12_{(3,0)}$+$R12_{(3,1)}$+$R12_{(3,2)}$+$R12_{(3,3)}$) >> 3

	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (0,3)	dc	dc	dc	dc
R15 (1,3)	dc	dc	dc	dc
R15 (2,3)	dc	dc	dc	dc
R15 (3,3)	dc	dc	dc	dc

4x4 Predicted block (DC)

	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (0,3)	R15 (0,3)	R15 (0,3)	R15 (0,3)	R15 (0,3)
R15 (1,3)	R15 (1,3)	R15 (1,3)	R15 (1,3)	R15 (1,3)
R15 (2,3)	R15 (2,3)	R15 (2,3)	R15 (2,3)	R15 (2,3)
R15 (3,3)	R15 (3,3)	R15 (3,3)	R15 (3,3)	R15 (3,3)

4x4 Predicted block (Horizontal)

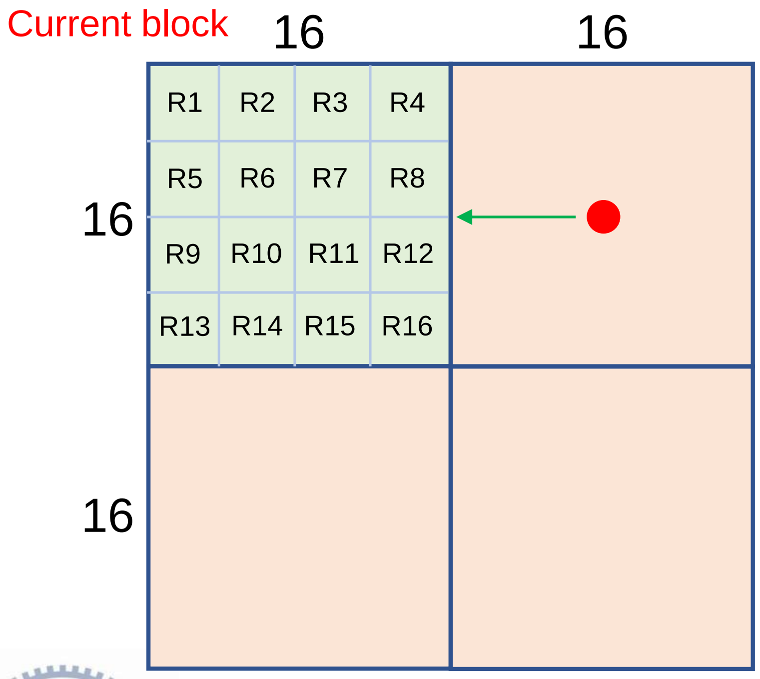
	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (0,3)	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (1,3)	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (2,3)	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)
R15 (3,3)	R12 (3,0)	R12 (3,1)	R12 (3,2)	R12 (3,3)

4x4 Predicted block (Vertical)

Case 2 – Intra 16x16 (1/10)

- Intra Prediction

Select the mode with the minimum SAD!



Input frame

Rx = Rx(0,3)+Rx(1,3)+Rx(2,3)+Rx(3,3) for x=1~16

dc = (R4+R8+R12+R16) >> 4

	N/A	N/A	N/A	N/A
(0,3) R4 (3,3)	dc	dc	dc	dc
(0,3) R8 (3,3)	dc	dc	dc	dc
(0,3) R12 (3,3)	dc	dc	dc	dc
(0,3) R16 (3,3)	dc	dc	dc	dc

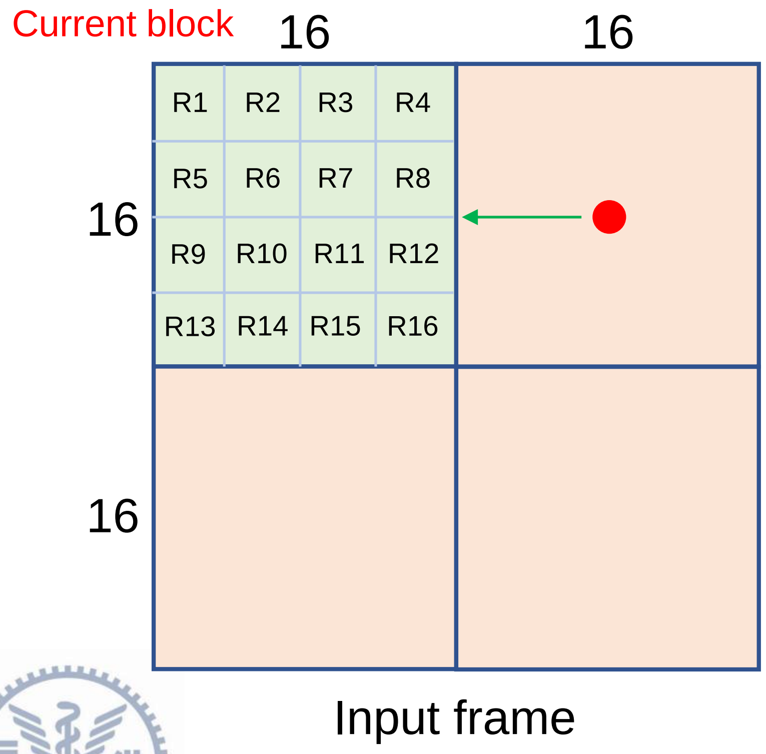
16x16 Predicted block (DC)

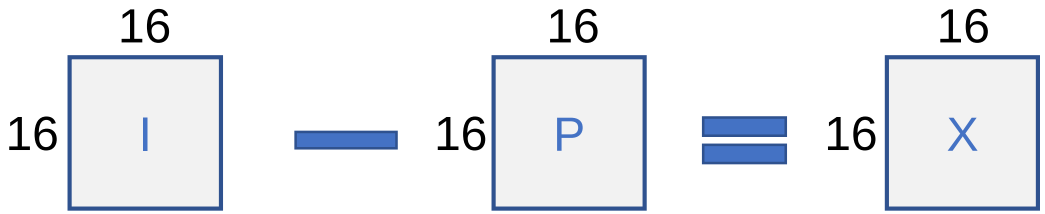
	N/A	N/A	N/A	N/A
(0,3) R4 (3,3)	(0,3) R4 (3,3)	(0,3) R4 (3,3)	(0,3) R4 (3,3)	(0,3) R4 (3,3)
(0,3) R8 (3,3)	(0,3) R8 (3,3)	(0,3) R8 (3,3)	(0,3) R8 (3,3)	(0,3) R8 (3,3)
(0,3) R12 (3,3)	(0,3) R12 (3,3)	(0,3) R12 (3,3)	(0,3) R12 (3,3)	(0,3) R12 (3,3)
(0,3) R16 (3,3)	(0,3) R16 (3,3)	(0,3) R16 (3,3)	(0,3) R16 (3,3)	(0,3) R16 (3,3)

16x16 Predicted block (Horizontal)

Case 2 – Intra 16x16 (2/10)

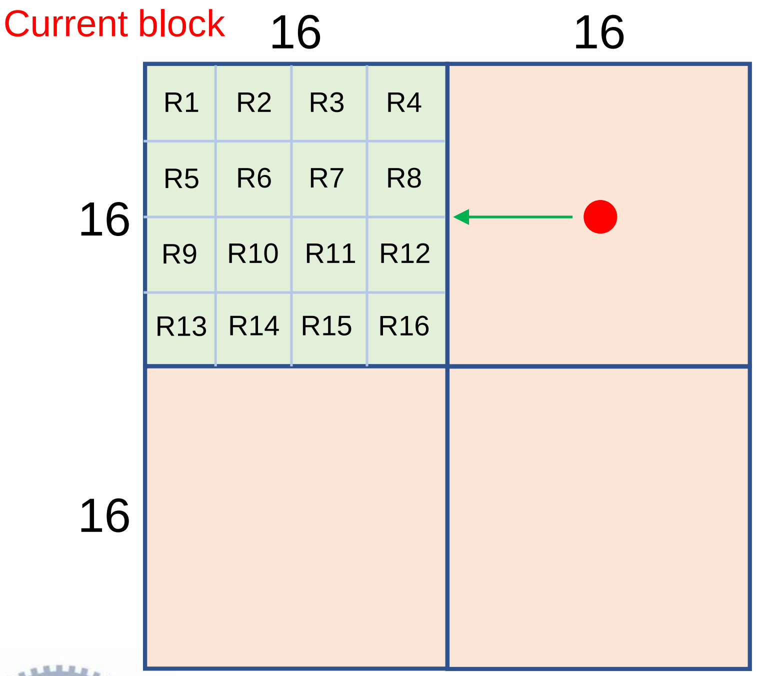
- Residual computation





Case 2 – Intra 16x16 (3/10)

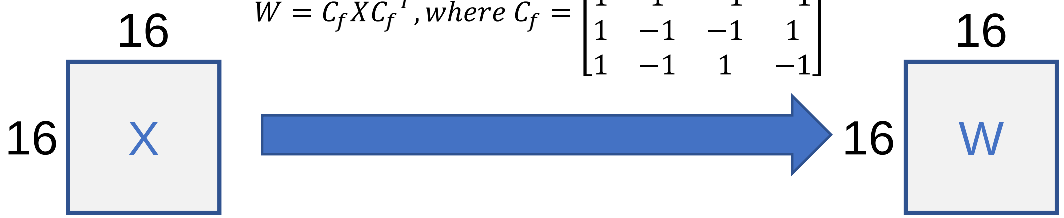
- Integer Transform



Input frame

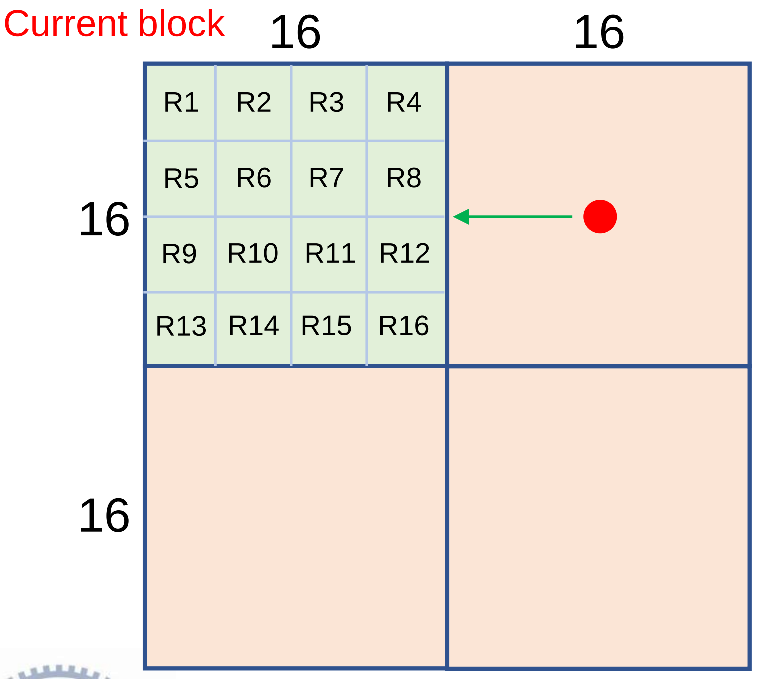
X is represented as 4x4 residual block

$$W = C_f X C_f^{T}, \text{where } C_f = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$



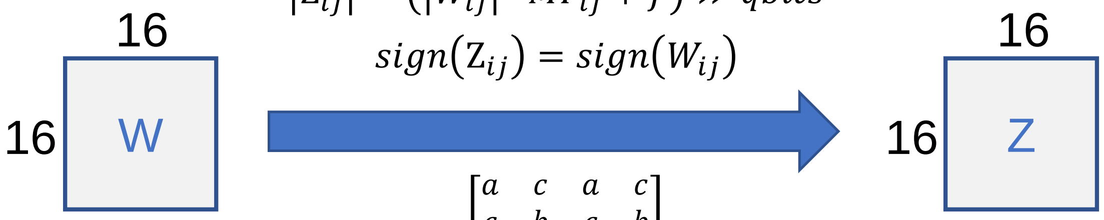
Case 2 – Intra 16x16 (4/10)

- Quantization



Input frame

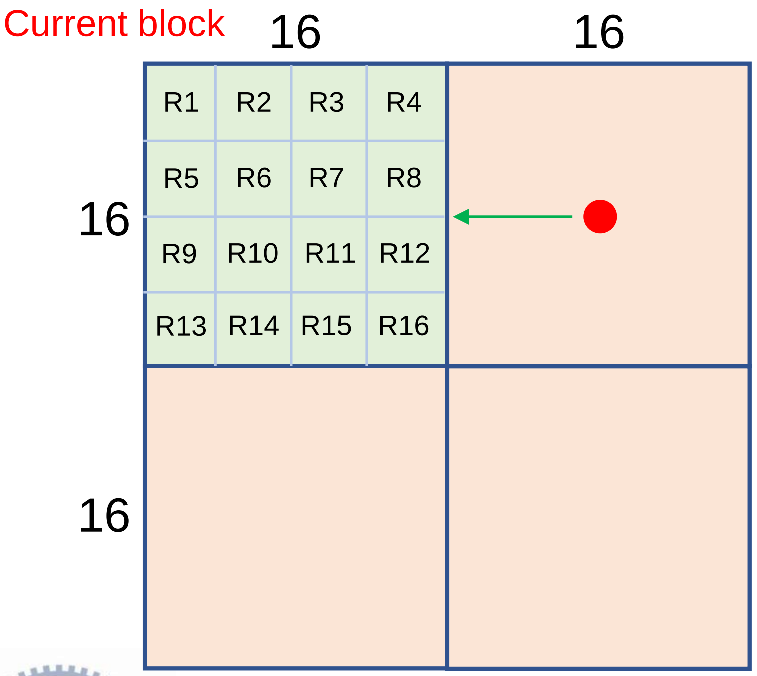
$$|Z_{ij}| = (|W_{ij}| \cdot MF_{ij} + f) \gg qbits$$
$$sign(Z_{ij}) = sign(W_{ij})$$



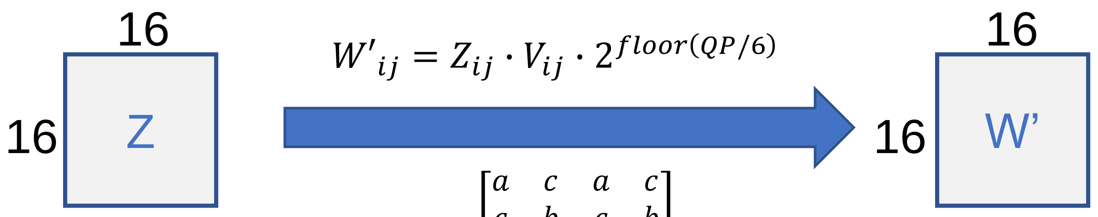
$$\mathrm{MF} = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

Case 2 – Intra 16x16 (5/10)

- Dequantization



Input frame

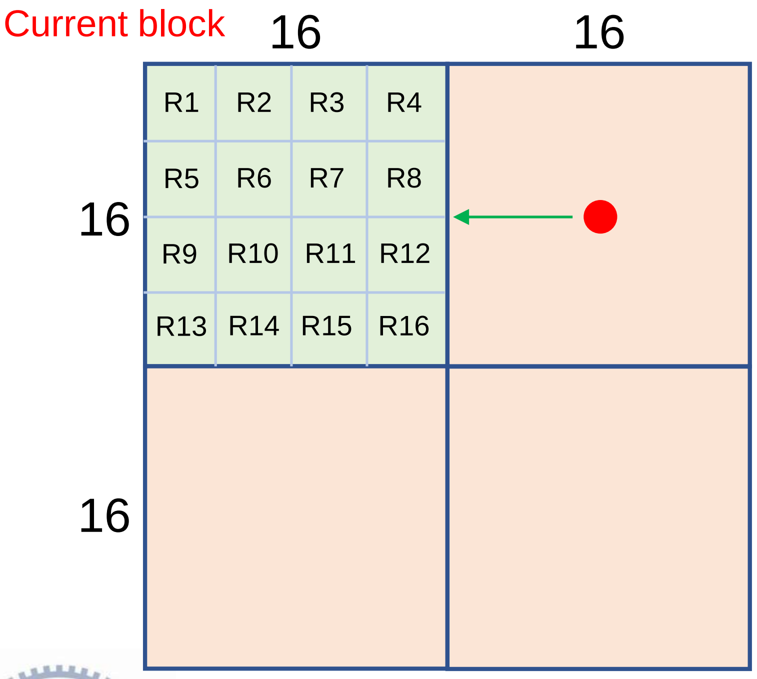


$$W'_{ij} = Z_{ij} \cdot V_{ij} \cdot 2^{floor(QP/6)}$$

$$V = \begin{bmatrix} a & c & a & c \\ c & b & c & b \\ a & c & a & c \\ c & b & c & b \end{bmatrix}$$

Case 2 – Intra 16x16 (6/10)

- Inverse Integer Transform

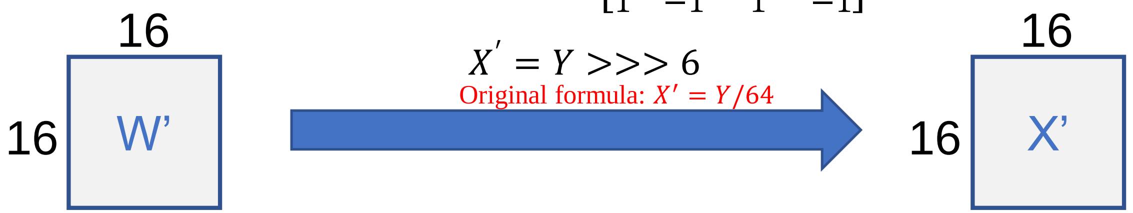


Input frame

$$Y = {C_i}^T W' C_i, where\ C_i = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}$$

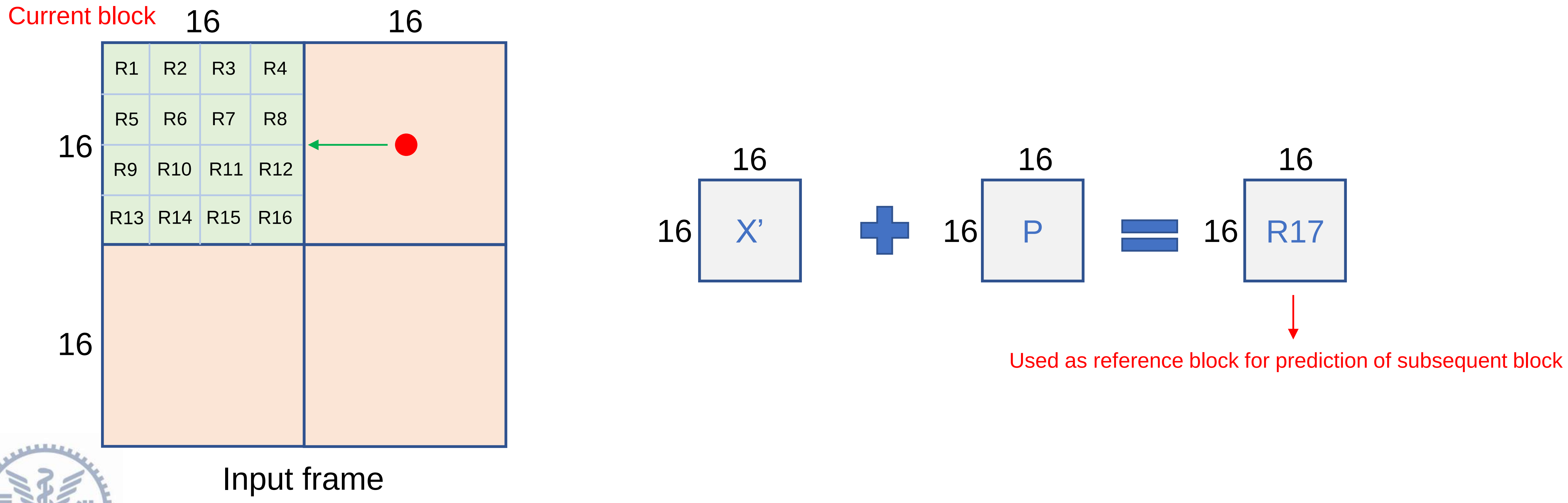
$$X' = Y >>> 6$$

Original formula: $X' = Y/64$



Case 2 – Intra 16x16 (7/10)

- Reconstruction



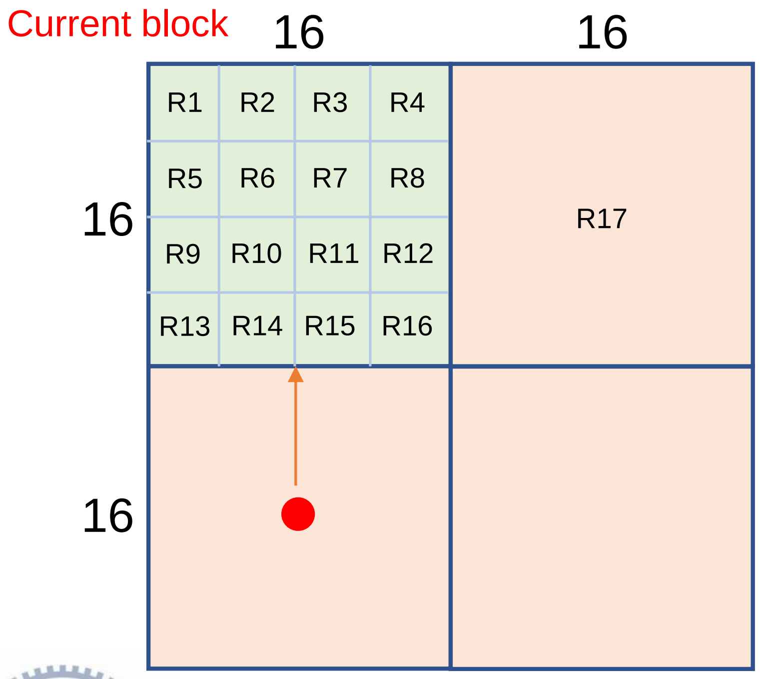
Case 2 – Intra 16x16 (8/10)

- Intra Prediction

Select the mode with the minimum SAD!

Rx = Rx(3,0)+Rx(3,1)+Rx(3,2)+Rx(3,3) for x=1~16

dc = (R13+R14+R15+R16) >> 4



Input frame

	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc
N/A	dc	dc	dc	dc

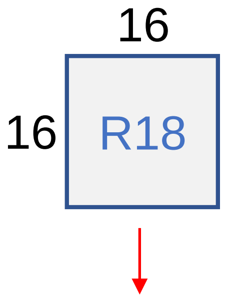
16x16 Predicted block (DC)

	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)
N/A	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)
N/A	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)
N/A	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)
N/A	R13 (3,0)(3,3)	R14 (3,0)(3,3)	R15 (3,0)(3,3)	R16 (3,0)(3,3)

16x16 Predicted block (Vertical)

Case 2 – Intra 16x16 (9/10)

- Residual computation
- Integer Transform
- Quantization
- Dequantization
- Inverse Integer Transform
- Reconstruction



Used as reference block for prediction of subsequent block

Case 2 – Intra 16x16 (10/10)

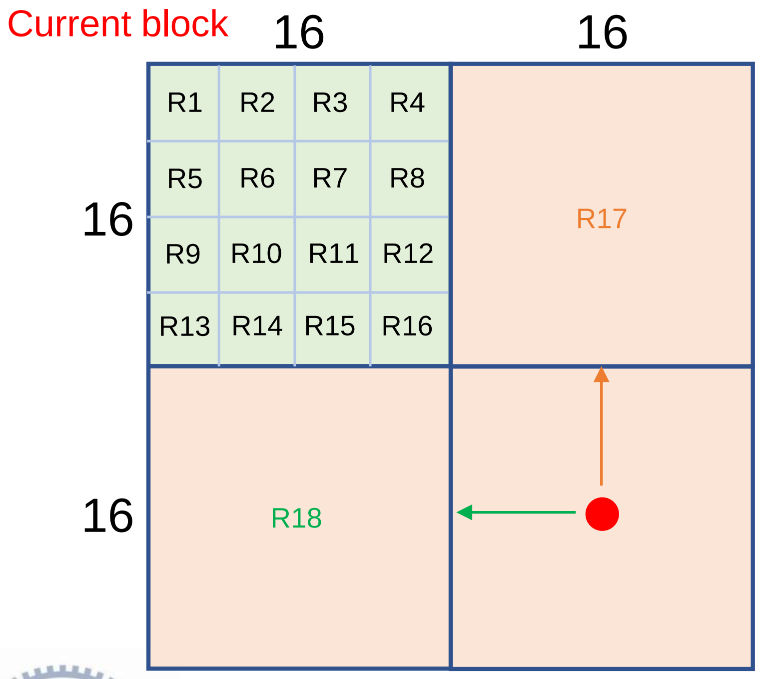
- Intra Prediction

Select the mode with the minimum SAD!

Rx = Rx(0,15)+Rx(1,15)+…+Rx(15,15) for x=1~16

Rx = Rx(15,0)+Rx(15,1)+…+Rx(15,15) for x=1~16

dc = (R17+R18) >> 5



Input frame

	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(0,15) R18 (3,15)	dc	dc	dc	dc
(4,15) R18 (7,15)	dc	dc	dc	dc
(8,15) R18 (11,15)	dc	dc	dc	dc
(12,15) R18 (15,5)	dc	dc	dc	dc

16x16 Predicted block (DC)

	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(0,15) R18 (3,15)	(0,15) R18 (3,15)	(0,15) R18 (3,15)	(0,15) R18 (3,15)	(0,15) R18 (3,15)
(4,15) R18 (7,15)	(4,15) R18 (7,15)	(4,15) R18 (7,15)	(4,15) R18 (7,15)	(4,15) R18 (7,15)
(8,15) R18 (11,15)	(8,15) R18 (11,15)	(8,15) R18 (11,15)	(8,15) R18 (11,15)	(8,15) R18 (11,15)
(12,15) R18 (15,5)	(12,15) R18 (15,5)	(12,15) R18 (15,5)	(12,15) R18 (15,5)	(12,15) R18 (15,5)

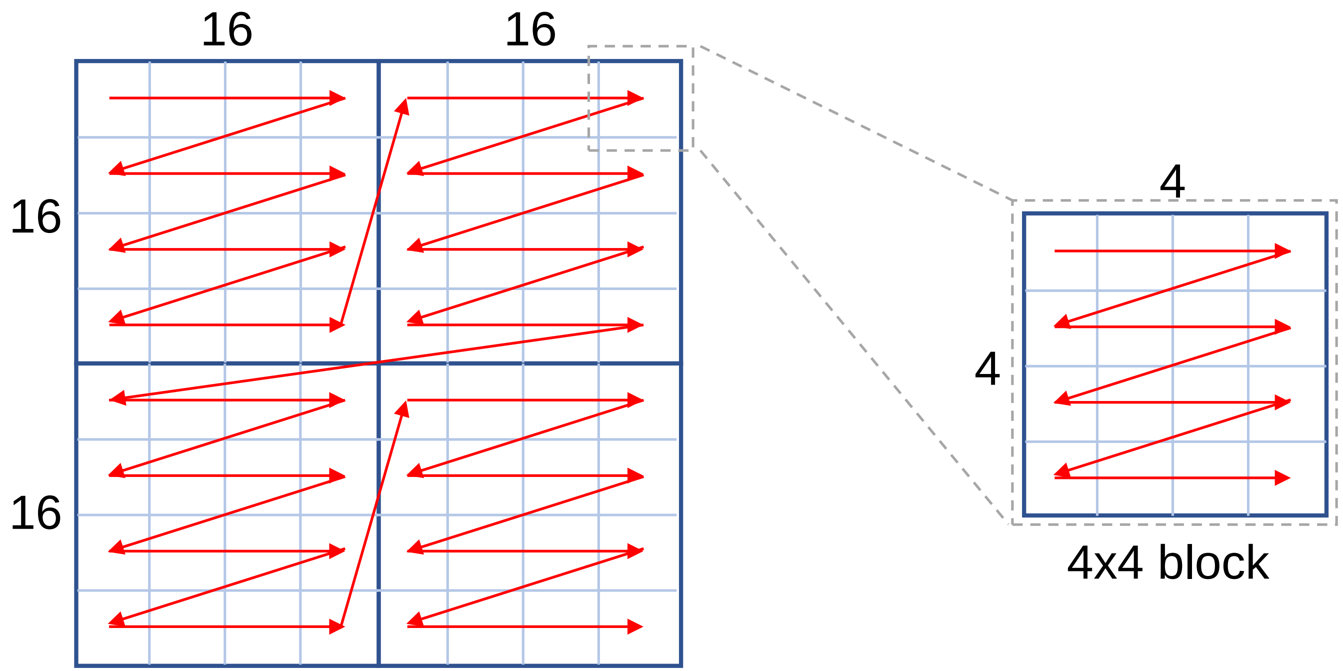
16x16 Predicted block (Horizontal)

	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(0,15) R18 (3,15)	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(4,15) R18 (7,15)	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(8,15) R18 (11,15)	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)
(12,15) R18 (15,5)	R17 (15,0) (15,3)	R17 (15,4) (15,7)	R17 (15,8) (15,11)	R17 (15,12) (15,15)

16x16 Predicted block (Vertical)

Output

- You should output the 32x32 Pre-entropy frame (Z) after HLPTE



Pre-entropy frame

Thanks